

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T10:03:07

Source: data/json/noise_experiments/v2/noise2i_full8match_game/claude-sonnet-4.5/game/noisy_bidirectional_informed_memprimary_full8match/noise2i_full8match_game_003_neutral_rep03.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary_full8match
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 3
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: self_and_coplayer
self_history_window: 0
coplayer_history_window: 3
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 3
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

ROUND 1

=====

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$5.5; payoffs Agent_1=\$7.5, Agent_2=\$3.5;

noise/visible: sent visible=\$3.8, received visible=\$11.4, returned visible=\$4.69

Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 9.0, "min": 0, "raw": 5.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 7.5,

"Agent_2": 3.5

}

Visible balances after round: {

"Agent_1": 6.69,

"Agent_2": 5.899999999999999

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:59:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance trust and caution since I don't have any history with the receiver yet.

Sending money creates an opportunity for mutual benefit since it gets tripled. If I send \$3:

- The receiver gets \$9 ($3 \times \3)
- I keep \$2 initially
- If they return a fair share, we both benefit

This shows willingness to cooperate while not overextending on the first round.

```
{'send': 3}
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual benefit
2. The receiver then decides how much to return
3. There's mentioned "communication noise" which means amounts may be displayed with some error

Key considerations:

- This is round 1, so I have no history with this receiver
- Sending \$0 guarantees I keep \$5, but misses opportunity for mutual gain
- Sending everything (\$5) maximizes potential but requires full trust
- A middle ground shows willingness to cooperate while limiting risk

Common approaches in trust games:

- Send about half (\$2-3) as a signal of trust while maintaining some security
- This allows testing the receiver's trustworthiness
- If they reciprocate fairly, we can build trust in future rounds

Given this is a multi-round game, I want to:

1. Signal cooperation
2. Test trustworthiness
3. Leave room to adjust strategy based on their response

I'll send \$3 as a reasonable middle-ground approach. This:

- Shows trust (sending more than half)
- Gives receiver \$9 to work with
- Leaves me with \$2 as a safety net
- Creates opportunity for mutual gain if receiver is fair

If the receiver returns a fair amount (ideally \$3-4 or more), it suggests they're cooperative and I can trust more in future rounds.

```
Usage: {  
  "input_tokens": 365,  
  "output_tokens": 449,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:59:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
}
```

```

"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.8, that is 76.0% of their total endowment. You received \$11.399999999999999. How much do you return to The sender? (0-11.399999999999999)

Assistant output:

Looking at this situation:

- The sender showed high trust by sending \$3.80 (76% of their endowment)
- I received \$11.40 after the 3x multiplier
- The sender currently has \$1.20 remaining

I want to reward this trust while ensuring we both benefit from cooperation. The sender took a risk by sending most of their endowment, so I should return enough to make it worthwhile for them.

If I return \$5.50:

- Sender ends with: \$1.20 + \$5.50 = \$6.70 (a gain of \$1.70)
- I keep: \$11.40 - \$5.50 = \$5.90 (a gain of \$0.90)

This gives the sender a good return on their trust while we both gain from cooperation.

{'return': 5.5}

Reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$3.8 (which is 76% of their \$5 endowment)
- I received \$11.40 (which is $3.8 \times 3 = 11.40$)

- I need to decide how much to return (between \$0 and \$11.40)

The sender showed significant trust by sending 76% of their endowment. Let me think about what would be fair:

- The sender kept: $\$5 - \$3.8 = \$1.2$
- I received: $\$11.40$
- Total created wealth: $\$1.2 + \$11.40 = \$12.60$

If I want to be fair and split the gains:

- We started with \$5 each (total \$10 if we consider the sender's initial endowment)
- After multiplication, we have \$12.60 total ($\$1.20 + \11.40)
- The gain from cooperation is $\$12.60 - \$5 = \$7.60$ (since the sender started with \$5)

One approach: Equal split of final amounts

- If we split \$12.60 equally: each gets \$6.30
- Sender currently has \$1.20, so I'd return $\$6.30 - \$1.20 = \$5.10$

Another approach: Reward the sender's trust

- The sender sent 76%, showing high trust
- I could return around 50% of what I received: $\$11.40 \times 0.5 = \5.70
- This would give sender: $\$1.20 + \$5.70 = \$6.90$
- I would keep: $\$11.40 - \$5.70 = \$5.70$
- Both end up with more than the initial \$5

Let me consider reciprocating proportionally:

- If I return 76% of \$11.40 = \$8.66
- This would give sender: $\$1.20 + \$8.66 = \$9.86$
- I would keep: $\$11.40 - \$8.66 = \$2.74$

That seems excessive. A more balanced approach would ensure the sender gains from their trust while I also benefit from receiving. Returning \$5-6 seems reasonable-it gives the sender a total of \$6.20-7.20 (more than their initial \$5) while I keep \$5.40-6.40.

Given the substantial...

```
Usage: {
  "input_tokens": 402,
  "output_tokens": 1024,
  "reasoning_tokens": 0
}
```

ROUND 2

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$3.8; received \$11.4; Agent_1 returned \$1; payoffs Agent_2=\$2.2, Agent_1=\$10.4; noise/visible: sent visible=\$0.83, received visible=\$2.49, returned visible=\$0.75
Action validation: {"returned": {"amount": 1.0, "clamped": false, "max": 11.399999999999999, "min": 0, "raw": 1.0}, "sent": {"amount": 3.8, "clamped": false, "max": 5, "min": 0, "raw": 3.8}}

Balances after round: {

"Agent_1": 17.9,
"Agent_2": 5.7

}

Visible balances after round: {

"Agent_1": 8.18,
"Agent_2": 7.849999999999999

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:59:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

```

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.8, that is 76.0% of their total endowment. You received \$11.399999999999999. How much do you return to The sender? (0-11.399999999999999)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed high trust by sending \$3.80 (76% of their endowment)
- I received \$11.40 after the 3x multiplier
- The sender currently has \$1.20 remaining

I want to reward this trust while ensuring we both benefit from cooperation. The sender took a risk by sending most of their endowment, so I should return enough to make it worth while for them.

If I return \$5.50:

- Sender ends with: $\$1.20 + \$5.50 = \$6.70$ (a gain of \$1.70)
- I keep: $\$11.40 - \$5.50 = \$5.90$ (a gain of \$0.90)

This gives the sender a good return on their trust while we both gain from cooperation.

```
{'return': 5.5}
```

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$3.8 (which is 76% of their \$5 endowment)
- I received \$11.40 (which is $\$3.8 \times 3 = \11.40)
- I need to decide how much to return (between \$0 and \$11.40)

The sender showed significant trust by sending 76% of their endowment. Let me think about what would be fair:

- The sender kept: $\$5 - \$3.8 = \$1.2$
- I received: \$11.40
- Total created wealth: $\$1.2 + \$11.40 = \$12.60$

If I want to be fair and split the gains:

- We started with \$5 each (total \$10 if we consider the sender's initial endowment)
- After multiplication, we have \$12.60 total ($\$1.20 + \11.40)
- The gain from cooperation is $\$12.60 - \$5 = \$7.60$ (since the sender started with \$5)

One approach: Equal split of final amounts

- If we split \$12.60 equally: each gets \$6.30
- Sender currently has \$1.20, so I'd return $\$6.30 - \$1.20 = \$5.10$

Another approach: Reward the sender's trust

- The sender sent 76%, showing high trust
- I could return around 50% of what I received: $\$11.40 \times 0.5 = \5.70
- This would give sender: $\$1.20 + \$5.70 = \$6.90$
- I would keep: $\$11.40 - \$5.70 = \$5.70$
- Both end up with more than the initial \$5

Let me consider reciprocating proportionally:

- If I return 76% of \$11.40 = \$8.66
- This would give sender: $\$1.20 + \$8.66 = \$9.86$
- I would keep: $\$11.40 - \$8.66 = \$2.74$

That seems excessive. A more balanced approach would ensure the sender gains from their trust while I also benefit from receiving. Returning \$5-6 seems reasonable-it gives the sender a total of \$6.20-7.20 (more than their initial \$5) while I keep \$5.40-6.40.

Given the substantial...

usage:

```
{
  "input_tokens": 402,
  "output_tokens": 1024,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.8999999999999999.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the previous round, the sender showed strong trust by sending \$3.80 (76% of their endowment), and I returned \$5.50 to reward that trust. This established a cooperative pattern.

I want to reciprocate this trust and continue the cooperative relationship. I'll send a similar proportion to demonstrate that I'm willing to trust as I was trusted.

****{'send': 3.8}****

Reasoning:

In Round 1, I was the receiver and the sender sent me \$3.80 (76% of their endowment), showing high trust. I returned \$5.50, which gave them a good return.

Now in Round 2, I'm the sender. I should consider:

1. The previous round showed that cooperation can work
2. The sender in Round 1 was generous, sending 76% of their endowment
3. I returned a fair amount that rewarded their trust
4. I should reciprocate this cooperative behavior

Since the previous sender sent \$3.80 (76%), I should consider sending a similar amount to maintain the cooperative relationship and signal trust.

My current total earnings are \$5.90 from Round 1.

If I send \$3.80:

- It shows similar trust to what was shown to me
- The receiver gets \$11.40 (3x multiplier)
- If they return proportionally (around 48% like I did), I'd get back around \$5.50
- I'd end up with $\$1.20 + \$5.50 = \$6.70$

However, I need to account for the note about communication noise - the amounts might differ from what's actually sent. This is important to keep in mind.

I think sending \$3.80 is a good choice to:

- Reciprocate the trust shown to me
- Continue building cooperation
- Signal that I'm willing to trust just as I was trusted

```
Usage: {  
  "input_tokens": 651,  
  "output_tokens": 429,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:00:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution since I don't have any history with the receiver yet.

Sending money creates an opportunity for mutual benefit since it gets tripled. If I send \$3:

- The receiver gets \$9 ($3 \times \3)
- I keep \$2 initially
- If they return a fair share, we both benefit

This shows willingness to cooperate while not overextending on the first round.

{'send': 3}

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual benefit
2. The receiver then decides how much to return
3. There's mentioned "communication noise" which means amounts may be displayed with some error

Key considerations:

- This is round 1, so I have no history with this receiver
- Sending \$0 guarantees I keep \$5, but misses opportunity for mutual gain
- Sending everything (\$5) maximizes potential but requires full trust
- A middle ground shows willingness to cooperate while limiting risk

Common approaches in trust games:

- Send about half (\$2-3) as a signal of trust while maintaining some security
- This allows testing the receiver's trustworthiness
- If they reciprocate fairly, we can build trust in future rounds

Given this is a multi-round game, I want to:

1. Signal cooperation
2. Test trustworthiness
3. Leave room to adjust strategy based on their response

I'll send \$3 as a reasonable middle-ground approach. This:

- Shows trust (sending more than half)
- Gives receiver \$9 to work with
- Leaves me with \$2 as a safety net
- Creates opportunity for mutual gain if receiver is fair

If the receiver returns a fair amount (ideally \$3-4 or more), it suggests they're cooperative and I can trust more in future rounds.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 449,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.69.

This round, The sender sent you \$0.83, so you received \$2.4899999999999998. How much do you return to The sender? (0-2.4899999999999998)

Assistant output:

Looking at this situation:

- The sender sent me \$0.83 (quite a small amount, showing limited trust)
- I received \$2.49 after tripling
- In Round 1, I sent \$3 and received a fairly generous return

Since the sender showed limited trust by only sending \$0.83, I'll return a moderate amount that's fair but proportional to their trust level. Returning about 40% of what I received seems reasonable - it gives them a small profit while I also benefit from the exchange.

```
{'return': 1.0}
```

Reasoning:

Let me analyze this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.83
3. I received \$2.49 (0.83×3)
4. I need to decide how much to return

Looking at my total earnings of \$6.69 across all rounds:

- Round 1: I sent \$3, so I kept \$2. The receiver would have gotten \$9.
- If the receiver returned something, that would add to my total
- My total is \$6.69, which means in Round 1, I got \$2 (what I kept) + some amount returned
- $\$6.69 - \$2 = \$4.69$ returned in Round 1

Wait, let me recalculate:

- Round 1: I sent \$3, kept \$2
- Receiver got \$9 (my $\$3 \times 3$)
- If receiver returned \$4.69, then my total from Round 1 = $\$2 + \$4.69 = \$6.69$

So the receiver returned \$4.69 out of \$9 they received, which is about 52% - that's fairly generous!

Now in Round 2:

- The sender sent me \$0.83 (very little)
- I received \$2.49
- The sender kept $\$5 - \$0.83 = \$4.17$

The sender was much less trusting/generous than I was in Round 1. They only sent \$0.83 compared to my \$3.

Given that:

1. The previous receiver was fairly generous to me (returned about 52%)
2. This sender was not very trusting (only sent \$0.83)

3. I should probably be fair but not overly generous to someone who didn't trust me much

A fair approach might be to return about 1/3 to 1/2 of what I received, similar to what was done for me.

Returning around \$1.25 seems reasonable-it gives the sender a modest gain of \$0.42 on their initial \$0.83 investment, while I keep \$1.24. This feels more balanced than returning just \$1.00, which would create too much disparity between what we each gain from the exchange.

I could also think about splitting the surplus created by the tripling. The \$2.49 represents a \$1.66 gain beyond the original \$0.83 investment. If I split that surplus equally, each of us would get our initial stake back plus half the gains, which would mean the sender should receive around \$1.66 total.

But they already have \$4.17 from their original \$5, so this framework doesn't quite work for determining what they should get back from me.

Looking at it more practically: they sent me \$0.83 and I now have \$2.49. A reciprocal approach would be returning roughly 50% of what I received, which comes to about \$1.25. I...

```
Usage: {
  "input_tokens": 550,
  "output_tokens": 1248,
  "reasoning_tokens": 0
}
```

ROUND 3

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$0; payoffs Agent_1=\$2, Agent_2=\$9;

noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.45

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 9.0, "min": 0, "raw": 0.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 19.9,

"Agent_2": 14.7

}

Visible balances after round: {

"Agent_1": 10.629999999999999,

"Agent_2": 7.849999999999999

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:00:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 3,

```
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution since I don't have any history with the receiver yet.

Sending money creates an opportunity for mutual benefit since it gets tripled. If I send \$3:

- The receiver gets \$9 (3 × \$3)
- I keep \$2 initially
- If they return a fair share, we both benefit

This shows willingness to cooperate while not overextending on the first round.

{'send': 3}

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual benefit
2. The receiver then decides how much to return
3. There's mentioned "communication noise" which means amounts may be displayed with some error

Key considerations:

- This is round 1, so I have no history with this receiver
- Sending \$0 guarantees I keep \$5, but misses opportunity for mutual gain
- Sending everything (\$5) maximizes potential but requires full trust

- A middle ground shows willingness to cooperate while limiting risk

Common approaches in trust games:

- Send about half (\$2-3) as a signal of trust while maintaining some security
- This allows testing the receiver's trustworthiness
- If they reciprocate fairly, we can build trust in future rounds

Given this is a multi-round game, I want to:

1. Signal cooperation
2. Test trustworthiness
3. Leave room to adjust strategy based on their response

I'll send \$3 as a reasonable middle-ground approach. This:

- Shows trust (sending more than half)
- Gives receiver \$9 to work with
- Leaves me with \$2 as a safety net
- Creates opportunity for mutual gain if receiver is fair

If the receiver returns a fair amount (ideally \$3-4 or more), it suggests they're cooperative and I can trust more in future rounds.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 449,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.69.

This round, The sender sent you \$0.83, so you received \$2.4899999999999998. How much do you return to The sender? (0-2.4899999999999998)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.83 (quite a small amount, showing limited trust)
- I received \$2.49 after tripling
- In Round 1, I sent \$3 and received a fairly generous return

Since the sender showed limited trust by only sending \$0.83, I'll return a moderate amount that's fair but proportional to their trust level. Returning about 40% of what I received seems reasonable - it gives them a small profit while I also benefit from the exchange.

```
{'return': 1.0}
```

reasoning:

Let me analyze this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.83
3. I received \$2.49 (0.83×3)
4. I need to decide how much to return

Looking at my total earnings of \$6.69 across all rounds:

- Round 1: I sent \$3, so I kept \$2. The receiver would have gotten \$9.
- If the receiver returned something, that would add to my total
- My total is \$6.69, which means in Round 1, I got \$2 (what I kept) + some amount returned
- $\$6.69 - \$2 = \$4.69$ returned in Round 1

Wait, let me recalculate:

- Round 1: I sent \$3, kept \$2
- Receiver got \$9 (my $\$3 \times 3$)
- If receiver returned \$4.69, then my total from Round 1 = $\$2 + \$4.69 = \$6.69$

So the receiver returned \$4.69 out of \$9 they received, which is about 52% - that's fairly generous!

Now in Round 2:

- The sender sent me \$0.83 (very little)
- I received \$2.49
- The sender kept \$5 - \$0.83 = \$4.17

The sender was much less trusting/generous than I was in Round 1. They only sent \$0.83 compared to my \$3.

Given that:

1. The previous receiver was fairly generous to me (returned about 52%)
2. This sender was not very trusting (only sent \$0.83)
3. I should probably be fair but not overly generous to someone who didn't trust me much

A fair approach might be to return about 1/3 to 1/2 of what I received, similar to what was done for me.

Returning around \$1.25 seems reasonable-it gives the sender a modest gain of \$0.42 on their initial \$0.83 investment, while I keep \$1.24. This feels more balanced than returning just \$1.00, which would create too much disparity between what we each gain from the exchange.

I could also think about splitting the surplus created by the tripling. The \$2.49 represents a \$1.66 gain beyond the original \$0.83 investment. If I split that surplus equally, each of us would get our initial stake back plus half the gains, which would mean the sender should receive around \$1.66 total.

But they already have \$4.17 from their original \$5, so this framework doesn't quite work for determining what they should get back from me.

Looking at it more practically: they sent me \$0.83 and I now have \$2.49. A reciprocal approach would be returning roughly 50% of what I received, which comes to about \$1.25. I...

usage:

```
{
  "input_tokens": 550,
  "output_tokens": 1248,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.18.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my experience so far:

Round 1 (as SENDER): I sent \$3 and received a very generous return, ending with \$6.69 total.

Round 2 (as RECEIVER): The sender only sent \$0.83 (showing caution), and I returned \$1.0, earning \$1.49 that round.

Now I have \$5 to allocate. Given that cooperation has worked well when both parties engaged (Round 1), I'll send a substantial amount again to encourage mutual benefit. The 3x multiplier means sending creates real value.

```
{'send': 3}
```

Reasoning:

Let me review what's happened so far:

Round 1 (I was SENDER):

- I sent \$3
- Receiver got \$9
- I started with \$5, sent \$3, kept \$2
- My total earnings are listed as \$6.69 after Round 2
- So I must have received something back in Round 1

Round 2 (I was RECEIVER):

- Sender sent me \$0.83
- I received \$2.49
- I returned \$1.0
- Sender kept \$5 - \$0.83 + \$1.0 = \$5.17
- I kept \$2.49 - \$1.0 = \$1.49

- My total after Round 2 was \$6.69

So after Round 1, I must have had $\$6.69 - \$1.49 = \$5.20$

If I kept \$2 from my initial \$5 and sent \$3, then I received \$3.20 back in Round 1.

Now Round 3:

- I'm the SENDER again
- My total earnings are \$8.18
- That means I earned $\$8.18 - \$6.69 = \$1.49$ in the interim (but wait, we're at the start of Round 3, so this should be my total after Round 2, which matches)

Actually, let me recalculate:

- After Round 1: I had some amount
- After Round 2: I had \$6.69
- Now Round 3 start: I have \$8.18

Wait, that doesn't make sense. If my total after Round 2 was \$6.69, and now at the start of Round 3 it's \$8.18, then I gained \$1.49 somehow. But Round 3 is just starting.

Let me re-read: "Your total earnings across all rounds are \$8.18"

Oh, I think this might include the \$5 I'm getting this round? So:

- After Round 2: \$6.69
- Round 3 starting money: \$5
- Total available: $\$6.69 + \$5 = \$11.69$

No, that's not right either. The \$1.49 gap is puzzling-unless "total earnings" means something different than I'm interpreting, or there's a mechanic I'm missing about how payoffs accumulate between rounds.

Let me reconsider what the prompts actually stated. The Round 2 message showed \$6.69 as my cumulative total at that point, meaning Round 1 netted me \$6.69. Then in Round 2, I received \$2.49 and returned \$1.0, keeping \$1.49, which brings me to \$8.18 for Round 3. This tracks-the \$1.49 gain matches the difference between the two totals.

For Round 3, I'm now the sender with \$5 available and \$8.18 in cumulative earnings so far. In Round 1 when I was the sender, I sent \$3 and apparently received enough back to reach that \$6.69 total.

Working backwards: I started with \$5, sent \$3 (keeping \$2), and ended Round 1 with \$6.69, so the return must have been \$4.69. That's remarkably generous-they received \$9 from my \$3 and sent back more than half of it. In Round 2, the sender was more cautious, giving me only \$2.49 from their \$5, though I reciprocated with a \$1.0 return....

```
Usage: {
  "input_tokens": 721,
  "output_tokens": 1303,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:00:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
}
```



```

"role": "trustee",
"role_label": "Receiver",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.8, that is 76.0% of their total endowment. You received \$11.399999999999999. How much do you return to The sender? (0-11.399999999999999)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed high trust by sending \$3.80 (76% of their endowment)
- I received \$11.40 after the 3x multiplier
- The sender currently has \$1.20 remaining

I want to reward this trust while ensuring we both benefit from cooperation. The sender took a risk by sending most of their endowment, so I should return enough to make it worth while for them.

If I return \$5.50:

- Sender ends with: $\$1.20 + \$5.50 = \$6.70$ (a gain of \$1.70)
- I keep: $\$11.40 - \$5.50 = \$5.90$ (a gain of \$0.90)

This gives the sender a good return on their trust while we both gain from cooperation.

{'return': 5.5}

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1

- The sender sent \$3.8 (which is 76% of their \$5 endowment)
- I received \$11.40 (which is $3.8 \times 3 = \$11.40$)
- I need to decide how much to return (between \$0 and \$11.40)

The sender showed significant trust by sending 76% of their endowment. Let me think about what would be fair:

- The sender kept: $\$5 - \$3.8 = \$1.2$
- I received: \$11.40
- Total created wealth: $\$1.2 + \$11.40 = \$12.60$

If I want to be fair and split the gains:

- We started with \$5 each (total \$10 if we consider the sender's initial endowment)
- After multiplication, we have \$12.60 total ($\$1.20 + \11.40)
- The gain from cooperation is $\$12.60 - \$5 = \$7.60$ (since the sender started with \$5)

One approach: Equal split of final amounts

- If we split \$12.60 equally: each gets \$6.30
- Sender currently has \$1.20, so I'd return $\$6.30 - \$1.20 = \$5.10$

Another approach: Reward the sender's trust

- The sender sent 76%, showing high trust
- I could return around 50% of what I received: $\$11.40 \times 0.5 = \5.70
- This would give sender: $\$1.20 + \$5.70 = \$6.90$
- I would keep: $\$11.40 - \$5.70 = \$5.70$
- Both end up with more than the initial \$5

Let me consider reciprocating proportionally:

- If I return 76% of \$11.40 = \$8.66
- This would give sender: $\$1.20 + \$8.66 = \$9.86$
- I would keep: $\$11.40 - \$8.66 = \$2.74$

That seems excessive. A more balanced approach would ensure the sender gains from their trust while I also benefit from receiving. Returning \$5-6 seems reasonable-it gives the sender a total of \$6.20-7.20 (more than their initial \$5) while I keep \$5.40-6.40.

Given the substantial...

usage:

```
{
  "input_tokens": 402,
  "output_tokens": 1024,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.8999999999999999.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender showed strong trust by sending \$3.80 (76% of their endowment), and I returned \$5.50 to reward that trust. This established a cooperative pattern.

I want to reciprocate this trust and continue the cooperative relationship. I'll send a similar proportion to demonstrate that I'm willing to trust as I was trusted.

```
**{'send': 3.8}**
```

reasoning:

In Round 1, I was the receiver and the sender sent me \$3.80 (76% of their endowment), showing high trust. I returned \$5.50, which gave them a good return.

Now in Round 2, I'm the sender. I should consider:

1. The previous round showed that cooperation can work
2. The sender in Round 1 was generous, sending 76% of their endowment
3. I returned a fair amount that rewarded their trust
4. I should reciprocate this cooperative behavior

Since the previous sender sent \$3.80 (76%), I should consider sending a similar amount to maintain the cooperative relationship and signal trust.

My current total earnings are \$5.90 from Round 1.

If I send \$3.80:

- It shows similar trust to what was shown to me
- The receiver gets \$11.40 (3x multiplier)
- If they return proportionally (around 48% like I did), I'd get back around \$5.50
- I'd end up with $\$1.20 + \$5.50 = \$6.70$

However, I need to account for the note about communication noise - the amounts might differ from what's actually sent. This is important to keep in mind.

I think sending \$3.80 is a good choice to:

- Reciprocate the trust shown to me
- Continue building cooperation
- Signal that I'm willing to trust just as I was trusted

usage:

```
{
  "input_tokens": 651,
  "output_tokens": 429,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.849999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0 and I received \$0, I have nothing to return.

This is unexpected given the cooperative pattern we established in the first two rounds. There may be communication noise affecting the game, or the sender's strategy changed.

****{'return': 0}****

Reasoning:

The sender sent me \$0 this round. This is interesting - in round 2, I sent \$3.8 (76% of my endowment), following the cooperative pattern from round 1. But now the sender sent me nothing.

There are several possible explanations:

1. The sender decided not to cooperate
2. There could be communication noise (the game mentioned this)
3. The sender might be testing me
4. Strategic behavior change

Since I received \$0, I can only return between \$0 and \$0. So I must return \$0.

This is a straightforward situation - I have nothing to return since I received nothing.

Usage: {

```
"input_tokens": 807,
"output_tokens": 214,
"reasoning_tokens": 0
}
```

ROUND 4

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$2.5; received \$7.5; Agent_1 returned \$0.4; payoffs Agent_2=\$2.9, Agent_1=\$7.1; noise/visible: sent visible=\$0.38, received visible=\$1.14, returned visible=\$0.25
Action validation: {"returned": {"amount": 0.4, "clamped": false, "max": 7.5, "min": 0, "raw": 0.4}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

```

"Agent_1": 27.0,
"Agent_2": 17.599999999999998
}
Visible balances after round: {
  "Agent_1": 11.37,
  "Agent_2": 10.599999999999998
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:01:00
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.8, that is 76.0% of their total endowment. You received \$11.399999999999999. How much do you return to The sender? (0-11.399999999999999)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed high trust by sending \$3.80 (76% of their endowment)
- I received \$11.40 after the 3x multiplier
- The sender currently has \$1.20 remaining

I want to reward this trust while ensuring we both benefit from cooperation. The sender took a risk by sending most of their endowment, so I should return enough to make it worthwhile for them.

If I return \$5.50:

- Sender ends with: $\$1.20 + \$5.50 = \$6.70$ (a gain of \$1.70)
- I keep: $\$11.40 - \$5.50 = \$5.90$ (a gain of \$0.90)

This gives the sender a good return on their trust while we both gain from cooperation.

{'return': 5.5}

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$3.8 (which is 76% of their \$5 endowment)
- I received \$11.40 (which is $\$3.8 \times 3 = \11.40)
- I need to decide how much to return (between \$0 and \$11.40)

The sender showed significant trust by sending 76% of their endowment. Let me think about what would be fair:

- The sender kept: $\$5 - \$3.8 = \$1.2$
- I received: \$11.40
- Total created wealth: $\$1.2 + \$11.40 = \$12.60$

If I want to be fair and split the gains:

- We started with \$5 each (total \$10 if we consider the sender's initial endowment)
- After multiplication, we have \$12.60 total ($\$1.20 + \11.40)
- The gain from cooperation is $\$12.60 - \$5 = \$7.60$ (since the sender started with \$5)

One approach: Equal split of final amounts

- If we split \$12.60 equally: each gets \$6.30
- Sender currently has \$1.20, so I'd return $\$6.30 - \$1.20 = \$5.10$

Another approach: Reward the sender's trust

- The sender sent 76%, showing high trust
- I could return around 50% of what I received: $\$11.40 \times 0.5 = \5.70
- This would give sender: $\$1.20 + \$5.70 = \$6.90$
- I would keep: $\$11.40 - \$5.70 = \$5.70$
- Both end up with more than the initial \$5

Let me consider reciprocating proportionally:

- If I return 76% of \$11.40 = \$8.66
- This would give sender: $\$1.20 + \$8.66 = \$9.86$
- I would keep: $\$11.40 - \$8.66 = \$2.74$

That seems excessive. A more balanced approach would ensure the sender gains from their trust while I also benefit from receiving. Returning \$5-6 seems reasonable-it gives the sender a total of \$6.20-7.20 (more than their initial \$5) while I keep \$5.40-6.40.

Given the substantial...

usage:

{

```
"input_tokens": 402,  
"output_tokens": 1024,  
"reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.8999999999999999.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender showed strong trust by sending \$3.80 (76% of the ir endowment), and I returned \$5.50 to reward that trust. This established a cooperative pattern.

I want to reciprocate this trust and continue the cooperative relationship. I'll send a similar proportion to demonstrate that I'm willing to trust as I was trusted.

****{'send': 3.8}****

reasoning:

In Round 1, I was the receiver and the sender sent me \$3.80 (76% of their endowment), showing high trust. I returned \$5.50, which gave them a good return.

Now in Round 2, I'm the sender. I should consider:

1. The previous round showed that cooperation can work
2. The sender in Round 1 was generous, sending 76% of their endowment
3. I returned a fair amount that rewarded their trust
4. I should reciprocate this cooperative behavior

Since the previous sender sent \$3.80 (76%), I should consider sending a similar amount to maintain the cooperative relationship and signal trust.

My current total earnings are \$5.90 from Round 1.

If I send \$3.80:

- It shows similar trust to what was shown to me
- The receiver gets \$11.40 (3x multiplier)
- If they return proportionally (around 48% like I did), I'd get back around \$5.50
- I'd end up with $\$1.20 + \$5.50 = \$6.70$

However, I need to account for the note about communication noise - the amounts might differ from what's actually sent. This is important to keep in mind.

I think sending \$3.80 is a good choice to:

- Reciprocate the trust shown to me
- Continue building cooperation
- Signal that I'm willing to trust just as I was trusted

usage:

```
{  
  "input_tokens": 651,  
  "output_tokens": 429,  
  "reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.8499999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0 and I received \$0, I have nothing to return.

This is unexpected given the cooperative pattern we established in the first two rounds. There may be communication noise affecting the game, or the sender's strategy changed.

```
**{'return': 0}**
```

reasoning:

The sender sent me \$0 this round. This is interesting - in round 2, I sent \$3.8 (76% of my endowment), following the cooperative pattern from round 1. But now the sender sent me nothing.

There are several possible explanations:

1. The sender decided not to cooperate
2. There could be communication noise (the game mentioned this)
3. The sender might be testing me
4. Strategic behavior change

Since I received \$0, I can only return between \$0 and \$0. So I must return \$0.

This is a straightforward situation - I have nothing to return since I received nothing.

usage:

```
{  
  "input_tokens": 807,  
  "output_tokens": 214,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.849999999999999.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my experience so far:

Round 2: I sent \$3.80 (showing high trust), but based on my earnings, I only received back a small amount (~\$0.75). This was disappointing.

Round 3: The sender sent me \$0.

However, the instructions mention there may be communication noise - what I observed might differ from what actually happened. This could explain the inconsistent behavior.

Given this uncertainty, I'll take a more moderate approach this round. I'll send \$2.50 (50% of my endowment) - enough to show willingness to cooperate and allow for mutual gains, but more cautious than my previous \$3.80 given the mixed signals.

```
**{'send': 2.5}**
```

Reasoning:

Let me think about what happened:

Round 1: I was RECEIVER

- Sender sent \$3.80 (76%)
- I received \$11.40
- I returned \$5.50
- My earnings: \$5.90

Round 2: I was SENDER

- I sent \$3.80
- My total earnings after this round were \$7.85
- So I gained $\$7.85 - \$5.90 = \$1.95$ from round 2
- If I sent \$3.80, I had \$1.20 left
- So I must have received back: $\$1.95 - \$1.20 = \$0.75$

Wait, that doesn't seem right. Let me recalculate.

Actually, my total earnings are cumulative. After round 1, I had \$5.90. After round 2, I have \$7.85. So in round 2, I gained $\$7.85 - \$5.90 = \$1.95$.

Round 2: I was SENDER

- I started with \$5
- I sent \$3.80
- I kept \$1.20
- Amount received back from receiver: must have been \$0.75

- Total from round 2: $\$1.20 + \$0.75 = \$1.95$?

So in round 2, I showed high trust (sending \$3.80), but the receiver only returned \$0.75. That's quite low cooperation from them.

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$7.85

The sender in round 3 sent nothing, possibly because they anticipated low cooperation based on their experience (or perhaps there was noise).

Now in Round 4, I'm the sender again.

The pattern is confusing. In my experience:

- Round 2: I sent \$3.80, got back only \$0.75 (low return)
- Round 3: The other player sent nothing, which could reflect communication noise or their own caution after previous rounds.

The instructions mention potential noise in the system, so what I'm observing might not be exactly what was intended. Given my mixed results-strong cooperation in round 1 but minimal returns when I took the risk in round 2-I need to balance caution with the possibility of mutual gain. If I send nothing, we each keep \$5, but if there's genuine willingness to cooperate, sending something could benefit us both....

```
Usage: {  
  "input_tokens": 918,  
  "output_tokens": 969,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:01:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution since I don't have any history with the receiver yet.

Sending money creates an opportunity for mutual benefit since it gets tripled. If I send \$3:

- The receiver gets \$9 ($3 \times \3)
- I keep \$2 initially
- If they return a fair share, we both benefit

This shows willingness to cooperate while not overextending on the first round.

{'send': 3}

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual benefit
2. The receiver then decides how much to return
3. There's mentioned "communication noise" which means amounts may be displayed with some error

Key considerations:

- This is round 1, so I have no history with this receiver
- Sending \$0 guarantees I keep \$5, but misses opportunity for mutual gain
- Sending everything (\$5) maximizes potential but requires full trust
- A middle ground shows willingness to cooperate while limiting risk

Common approaches in trust games:

- Send about half (\$2-3) as a signal of trust while maintaining some security
- This allows testing the receiver's trustworthiness
- If they reciprocate fairly, we can build trust in future rounds

Given this is a multi-round game, I want to:

1. Signal cooperation
2. Test trustworthiness
3. Leave room to adjust strategy based on their response

I'll send \$3 as a reasonable middle-ground approach. This:

- Shows trust (sending more than half)

- Gives receiver \$9 to work with
- Leaves me with \$2 as a safety net
- Creates opportunity for mutual gain if receiver is fair

If the receiver returns a fair amount (ideally \$3-4 or more), it suggests they're cooperative and I can trust more in future rounds.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 449,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.69.

This round, The sender sent you \$0.83, so you received \$2.4899999999999998. How much do you return to The sender? (0-2.4899999999999998)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.83 (quite a small amount, showing limited trust)
- I received \$2.49 after tripling
- In Round 1, I sent \$3 and received a fairly generous return

Since the sender showed limited trust by only sending \$0.83, I'll return a moderate amount that's fair but proportional to their trust level. Returning about 40% of what I received seems reasonable - it gives them a small profit while I also benefit from the exchange.

```
{'return': 1.0}
```

reasoning:

Let me analyze this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.83
3. I received \$2.49 (0.83×3)
4. I need to decide how much to return

Looking at my total earnings of \$6.69 across all rounds:

- Round 1: I sent \$3, so I kept \$2. The receiver would have gotten \$9.
- If the receiver returned something, that would add to my total
- My total is \$6.69, which means in Round 1, I got \$2 (what I kept) + some amount returned
- $\$6.69 - \$2 = \$4.69$ returned in Round 1

Wait, let me recalculate:

- Round 1: I sent \$3, kept \$2
- Receiver got \$9 (my $\$3 \times 3$)
- If receiver returned \$4.69, then my total from Round 1 = $\$2 + \$4.69 = \$6.69$

So the receiver returned \$4.69 out of \$9 they received, which is about 52% - that's fairly generous!

Now in Round 2:

- The sender sent me \$0.83 (very little)
- I received \$2.49
- The sender kept \$5 - $\$0.83 = \4.17

The sender was much less trusting/generous than I was in Round 1. They only sent \$0.83 compared to my \$3.

Given that:

1. The previous receiver was fairly generous to me (returned about 52%)
2. This sender was not very trusting (only sent \$0.83)
3. I should probably be fair but not overly generous to someone who didn't trust me much

A fair approach might be to return about 1/3 to 1/2 of what I received, similar to what was done for me.

Returning around \$1.25 seems reasonable-it gives the sender a modest gain of \$0.42 on their initial \$0.83 investment, while I keep \$1.24. This feels more balanced than returning just \$1.00, which would create too much disparity between what we each gain from the exchange.

I could also think about splitting the surplus created by the tripling. The \$2.49 represents a \$1.66 gain beyond the original \$0.83 investment. If I split that surplus equally, each of us would get our initial stake back plus half the gains, which would mean the sender should receive around \$1.66 total.

But they already have \$4.17 from their original \$5, so this framework doesn't quite work for determining what they should get back from me.

Looking at it more practically: they sent me \$0.83 and I now have \$2.49. A reciprocal approach would be returning roughly 50% of what I received, which comes to about \$1.25. I...

usage:

```
{
  "input_tokens": 550,
  "output_tokens": 1248,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.18.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my experience so far:

Round 1 (as SENDER): I sent \$3 and received a very generous return, ending with \$6.69 total.

Round 2 (as RECEIVER): The sender only sent \$0.83 (showing caution), and I returned \$1.0, earning \$1.49 that round.

Now I have \$5 to allocate. Given that cooperation has worked well when both parties engaged (Round 1), I'll send a substantial amount again to encourage mutual benefit. The 3x multiplier means sending creates real value.

{'send': 3}

reasoning:

Let me review what's happened so far:

Round 1 (I was SENDER):

- I sent \$3
- Receiver got \$9
- I started with \$5, sent \$3, kept \$2
- My total earnings are listed as \$6.69 after Round 2
- So I must have received something back in Round 1

Round 2 (I was RECEIVER):

- Sender sent me \$0.83
- I received \$2.49
- I returned \$1.0
- Sender kept $\$5 - \$0.83 + \$1.0 = \5.17
- I kept $\$2.49 - \$1.0 = \$1.49$
- My total after Round 2 was \$6.69

So after Round 1, I must have had $\$6.69 - \$1.49 = \$5.20$

If I kept \$2 from my initial \$5 and sent \$3, then I received \$3.20 back in Round 1.

Now Round 3:

- I'm the SENDER again
- My total earnings are \$8.18
- That means I earned $\$8.18 - \$6.69 = \$1.49$ in the interim (but wait, we're at the start of Round 3, so this should be my total after Round 2, which matches)

Actually, let me recalculate:

- After Round 1: I had some amount
- After Round 2: I had \$6.69
- Now Round 3 start: I have \$8.18

Wait, that doesn't make sense. If my total after Round 2 was \$6.69, and now at the start of Round 3 it's \$8.18, then I gained \$1.49 somehow. But Round 3 is just starting.

Let me re-read: "Your total earnings across all rounds are \$8.18"

Oh, I think this might include the \$5 I'm getting this round? So:

- After Round 2: \$6.69
- Round 3 starting money: \$5
- Total available: \$6.69 + \$5 = \$11.69

No, that's not right either. The \$1.49 gap is puzzling-unless "total earnings" means something different than I'm interpreting, or there's a mechanic I'm missing about how payoff accumulates between rounds.

Let me reconsider what the prompts actually stated. The Round 2 message showed \$6.69 as my cumulative total at that point, meaning Round 1 netted me \$6.69. Then in Round 2, I received \$2.49 and returned \$1.0, keeping \$1.49, which brings me to \$8.18 for Round 3. This tracks-the \$1.49 gain matches the difference between the two totals.

For Round 3, I'm now the sender with \$5 available and \$8.18 in cumulative earnings so far. In Round 1 when I was the sender, I sent \$3 and apparently received enough back to reach that \$6.69 total.

Working backwards: I started with \$5, sent \$3 (keeping \$2), and ended Round 1 with \$6.69, so the return must have been \$4.69. That's remarkably generous-they received \$9 from my \$3 and sent back more than half of it. In Round 2, the sender was more cautious, giving me only \$2.49 from their \$5, though I reciprocated with a \$1.0 return....

usage:

```
{
  "input_tokens": 721,
  "output_tokens": 1303,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.629999999999999.

This round, The sender sent you \$0.38, so you received \$1.1400000000000001. How much do you return to The sender? (0-1.1400000000000001)

Assistant output:

Looking at this round:

- The sender sent me only \$0.38 (very low amount, showing minimal trust)
- I received \$1.14 after tripling

In Round 3, I sent \$3 generously and received back very little in return (only about \$0.45 from \$9 given to the receiver).

This current sender is showing very limited trust with just \$0.38. Given:

1. Their low initial trust
2. The pattern of low returns I've been experiencing
3. The need to be proportionally fair

I'll return a modest amount - about 35% of what I received, which still gives them a small gain while protecting myself.

```
{'return': 0.4}
```

Reasoning:

Let me analyze this situation:

Round 3 (just completed as SENDER): I sent \$3, and my total went from \$8.18 to \$10.63. That means I gained \$2.45 that round.

- I started Round 3 with \$5
- I sent \$3, so I had \$2 left

- I ended with total of \$10.63, which was an increase of \$2.45 from \$8.18
- So I received back about \$0.45 from the receiver

Wait, that seems odd. Let me recalculate:

- Before Round 3: \$8.18 total
- Round 3 start: I get \$5 new
- Round 3: I send \$3, keep \$2
- After Round 3: \$10.63 total
- So from Round 3 I gained: $\$10.63 - \$8.18 = \$2.45$
- I kept \$2 from not sending, so I must have received back \$0.45

The receiver got \$9 ($3 \times \3) and only returned \$0.45? That's quite unfair - only 5% return rate!

Now Round 4 (as RECEIVER):

- The sender sent me only \$0.38 (very low trust!)
- I received \$1.14
- Given the previous receiver was very stingy with me, and this sender is showing very low trust...

However, I should consider:

1. This might be a different receiver in Round 3 vs this sender in Round 4
2. There's "communication noise" mentioned - amounts may differ from what was actually sent/received
3. I should still try to be fair and build cooperation

The sender sent very little (\$0.38), showing minimal trust. I could:

- Return nothing (playing it safe since they showed little trust)
- Return around 30-40% (\$0.34-\$0.46) which is proportionally fair

...

```
Usage: {
  "input_tokens": 938,
  "output_tokens": 720,
  "reasoning_tokens": 0
}
```

ROUND 5

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$2; received \$6; Agent_2 returned \$0.6; payoffs Agent_1=\$3.6, Agent_2=\$5.4;
 noise/visible: sent visible=\$0.42, received visible=\$1.26, returned visible=\$1.16
 Action validation: {"returned": {"amount": 0.6, "clamped": false, "max": 6.0, "min": 0, "raw": 0.6}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 30.6,
 "Agent_2": 23.0

}

Visible balances after round: {

"Agent_1": 15.53,
 "Agent_2": 11.259999999999998

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:01:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_2",
 "pairing": {
 "agents": [
 "Agent_1",

```

    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution since I don't have any history with the receiver yet.

Sending money creates an opportunity for mutual benefit since it gets tripled. If I send \$3:

- The receiver gets \$9 (3 × \$3)
- I keep \$2 initially
- If they return a fair share, we both benefit

This shows willingness to cooperate while not overextending on the first round.

{'send': 3}

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual benefit
2. The receiver then decides how much to return
3. There's mentioned "communication noise" which means amounts may be displayed with some error

Key considerations:

- This is round 1, so I have no history with this receiver
- Sending \$0 guarantees I keep \$5, but misses opportunity for mutual gain
- Sending everything (\$5) maximizes potential but requires full trust
- A middle ground shows willingness to cooperate while limiting risk

Common approaches in trust games:

- Send about half (\$2-3) as a signal of trust while maintaining some security
- This allows testing the receiver's trustworthiness
- If they reciprocate fairly, we can build trust in future rounds

Given this is a multi-round game, I want to:

1. Signal cooperation
2. Test trustworthiness
3. Leave room to adjust strategy based on their response

I'll send \$3 as a reasonable middle-ground approach. This:

- Shows trust (sending more than half)
- Gives receiver \$9 to work with
- Leaves me with \$2 as a safety net
- Creates opportunity for mutual gain if receiver is fair

If the receiver returns a fair amount (ideally \$3-4 or more), it suggests they're cooperative and I can trust more in future rounds.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 449,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.69.

This round, The sender sent you \$0.83, so you received \$2.4899999999999998. How much do you return to The sender? (0-2.4899999999999998)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.83 (quite a small amount, showing limited trust)
- I received \$2.49 after tripling
- In Round 1, I sent \$3 and received a fairly generous return

Since the sender showed limited trust by only sending \$0.83, I'll return a moderate amount that's fair but proportional to their trust level. Returning about 40% of what I received seems reasonable - it gives them a small profit while I also benefit from the exchange.

```
{'return': 1.0}
```

reasoning:

Let me analyze this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.83
3. I received \$2.49 (0.83×3)
4. I need to decide how much to return

Looking at my total earnings of \$6.69 across all rounds:

- Round 1: I sent \$3, so I kept \$2. The receiver would have gotten \$9.
- If the receiver returned something, that would add to my total

- My total is \$6.69, which means in Round 1, I got \$2 (what I kept) + some amount returned
- $\$6.69 - \$2 = \$4.69$ returned in Round 1

Wait, let me recalculate:

- Round 1: I sent \$3, kept \$2
- Receiver got \$9 (my $\$3 \times 3$)
- If receiver returned \$4.69, then my total from Round 1 = $\$2 + \$4.69 = \$6.69$

So the receiver returned \$4.69 out of \$9 they received, which is about 52% - that's fairly generous!

Now in Round 2:

- The sender sent me \$0.83 (very little)
- I received \$2.49
- The sender kept $\$5 - \$0.83 = \$4.17$

The sender was much less trusting/generous than I was in Round 1. They only sent \$0.83 compared to my \$3.

Given that:

1. The previous receiver was fairly generous to me (returned about 52%)
2. This sender was not very trusting (only sent \$0.83)
3. I should probably be fair but not overly generous to someone who didn't trust me much

A fair approach might be to return about 1/3 to 1/2 of what I received, similar to what was done for me.

Returning around \$1.25 seems reasonable-it gives the sender a modest gain of \$0.42 on their initial \$0.83 investment, while I keep \$1.24. This feels more balanced than returning just \$1.00, which would create too much disparity between what we each gain from the exchange.

I could also think about splitting the surplus created by the tripling. The \$2.49 represents a \$1.66 gain beyond the original \$0.83 investment. If I split that surplus equally, each of us would get our initial stake back plus half the gains, which would mean the sender should receive around \$1.66 total.

But they already have \$4.17 from their original \$5, so this framework doesn't quite work for determining what they should get back from me.

Looking at it more practically: they sent me \$0.83 and I now have \$2.49. A reciprocal approach would be returning roughly 50% of what I received, which comes to about \$1.25. I...

usage:

```
{
  "input_tokens": 550,
  "output_tokens": 1248,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.18.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my experience so far:

Round 1 (as SENDER): I sent \$3 and received a very generous return, ending with \$6.69 total.

Round 2 (as RECEIVER): The sender only sent \$0.83 (showing caution), and I returned \$1.00, earning \$1.49 that round.

Now I have \$5 to allocate. Given that cooperation has worked well when both parties engaged (Round 1), I'll send a substantial amount again to encourage mutual benefit. The 3x multiplier means sending creates real value.

```
{'send': 3}
```

reasoning:

Let me review what's happened so far:

Round 1 (I was SENDER):

- I sent \$3
- Receiver got \$9
- I started with \$5, sent \$3, kept \$2
- My total earnings are listed as \$6.69 after Round 2
- So I must have received something back in Round 1

Round 2 (I was RECEIVER):

- Sender sent me \$0.83
- I received \$2.49
- I returned \$1.0
- Sender kept $\$5 - \$0.83 + \$1.0 = \5.17
- I kept $\$2.49 - \$1.0 = \$1.49$
- My total after Round 2 was \$6.69

So after Round 1, I must have had $\$6.69 - \$1.49 = \$5.20$

If I kept \$2 from my initial \$5 and sent \$3, then I received \$3.20 back in Round 1.

Now Round 3:

- I'm the SENDER again
- My total earnings are \$8.18
- That means I earned $\$8.18 - \$6.69 = \$1.49$ in the interim (but wait, we're at the start of Round 3, so this should be my total after Round 2, which matches)

Actually, let me recalculate:

- After Round 1: I had some amount
- After Round 2: I had \$6.69
- Now Round 3 start: I have \$8.18

Wait, that doesn't make sense. If my total after Round 2 was \$6.69, and now at the start of Round 3 it's \$8.18, then I gained \$1.49 somehow. But Round 3 is just starting.

Let me re-read: "Your total earnings across all rounds are \$8.18"

Oh, I think this might include the \$5 I'm getting this round? So:

- After Round 2: \$6.69
- Round 3 starting money: \$5
- Total available: $\$6.69 + \$5 = \$11.69$

No, that's not right either. The \$1.49 gap is puzzling-unless "total earnings" means something different than I'm interpreting, or there's a mechanic I'm missing about how payoffs accumulate between rounds.

Let me reconsider what the prompts actually stated. The Round 2 message showed \$6.69 as my cumulative total at that point, meaning Round 1 netted me \$6.69. Then in Round 2, I received \$2.49 and returned \$1.0, keeping \$1.49, which brings me to \$8.18 for Round 3. This tracks-the \$1.49 gain matches the difference between the two totals.

For Round 3, I'm now the sender with \$5 available and \$8.18 in cumulative earnings so far. In Round 1 when I was the sender, I sent \$3 and apparently received enough back to reach that \$6.69 total.

Working backwards: I started with \$5, sent \$3 (keeping \$2), and ended Round 1 with \$6.69, so the return must have been \$4.69. That's remarkably generous-they received \$9 from my \$3 and sent back more than half of it. In Round 2, the sender was more cautious, giving me only \$2.49 from their \$5, though I reciprocated with a \$1.0 return....

usage:

```
{
  "input_tokens": 721,
  "output_tokens": 1303,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.629999999999999.

This round, The sender sent you \$0.38, so you received \$1.1400000000000001. How much do you

u return to The sender? (0-1.1400000000000001)

Message 9 role=assistant

content:

Looking at this round:

- The sender sent me only \$0.38 (very low amount, showing minimal trust)
- I received \$1.14 after tripling

In Round 3, I sent \$3 generously and received back very little in return (only about \$0.45 from \$9 given to the receiver).

This current sender is showing very limited trust with just \$0.38. Given:

1. Their low initial trust
2. The pattern of low returns I've been experiencing
3. The need to be proportionally fair

I'll return a modest amount - about 35% of what I received, which still gives them a small gain while protecting myself.

{'return': 0.4}

reasoning:

Let me analyze this situation:

Round 3 (just completed as SENDER): I sent \$3, and my total went from \$8.18 to \$10.63. That means I gained \$2.45 that round.

- I started Round 3 with \$5
- I sent \$3, so I had \$2 left
- I ended with total of \$10.63, which was an increase of \$2.45 from \$8.18
- So I received back about \$0.45 from the receiver

Wait, that seems odd. Let me recalculate:

- Before Round 3: \$8.18 total
- Round 3 start: I get \$5 new
- Round 3: I send \$3, keep \$2
- After Round 3: \$10.63 total
- So from Round 3 I gained: \$10.63 - \$8.18 = \$2.45
- I kept \$2 from not sending, so I must have received back \$0.45

The receiver got \$9 (3 × \$3) and only returned \$0.45? That's quite unfair - only 5% return rate!

Now Round 4 (as RECEIVER):

- The sender sent me only \$0.38 (very low trust!)
- I received \$1.14
- Given the previous receiver was very stingy with me, and this sender is showing very low trust...

However, I should consider:

1. This might be a different receiver in Round 3 vs this sender in Round 4
2. There's "communication noise" mentioned - amounts may differ from what was actually sent/returned
3. I should still try to be fair and build cooperation

The sender sent very little (\$0.38), showing minimal trust. I could:

- Return nothing (playing it safe since they showed little trust)
- Return around 30-40% (\$0.34-\$0.46) which is proportionally fair

...

usage:

```
{
  "input_tokens": 938,
  "output_tokens": 720,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.37.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my history:

Round 1 (SENDER): Sent \$3, received good return
Round 2 (RECEIVER): Received \$2.49, returned \$1.0
Round 3 (SENDER): Sent \$3, received very little back (~\$0.45)
Round 4 (RECEIVER): Received \$1.14, returned \$0.4

The pattern shows mixed results. When I was generous in Round 3, I got very little return. The receivers vary significantly in their reciprocity.

I'll send a moderate amount this round - showing willingness to cooperate but being somewhat cautious after the disappointing Round 3 experience.

{'send': 2}

Reasoning:

Let me review my history:

Round 1 (SENDER): Sent \$3, received generous return, ended with \$6.69 total
Round 2 (RECEIVER): Got \$2.49, returned \$1.0, earned \$1.49 that round ? \$8.18 total
Round 3 (SENDER): Sent \$3, but received only ~\$0.45 back, earned \$2.45 ? \$10.63 total
Round 4 (RECEIVER): Got \$1.14, returned \$0.4, earned \$0.74 ? \$11.37 total

So when I've been the sender and sent \$3:

- Round 1: Got a good return
- Round 3: Got very little back (~\$0.45 from the \$9 I gave them)

The receivers I'm dealing with seem to vary widely. In Round 3 I was generous and got very little back. In Round 4, someone sent me only \$0.38.

There's a note about communication noise - amounts may differ slightly from what was actually sent/returned. This could explain some discrepancies.

For Round 5, I should consider:

1. Being the sender twice before, I got mixed results
2. The last time I sent \$3 (Round 3), I got very little back
3. But the first time (Round 1) worked well
4. I might be dealing with a different person each round

Given the mixed results and the fact that last time I was generous as a sender I got burned, maybe I should be more cautious. But cooperation is still the best strategy overall if both parties engage.

I could:

- Send \$2 (moderate amount, less risk)
- Send \$3 (same as before, show consistency)
- Send less (be more cautious after Round 3)

...

```
Usage: {  
  "input_tokens": 1145,  
  "output_tokens": 627,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:01:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",
```

```

    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.8, that is 76.0% of their total endowment. You received \$11.399999999999999. How much do you return to The sender? (0-11.399999999999999)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed high trust by sending \$3.80 (76% of their endowment)
- I received \$11.40 after the 3x multiplier
- The sender currently has \$1.20 remaining

I want to reward this trust while ensuring we both benefit from cooperation. The sender took a risk by sending most of their endowment, so I should return enough to make it worthwhile for them.

If I return \$5.50:

- Sender ends with: $\$1.20 + \$5.50 = \$6.70$ (a gain of \$1.70)
- I keep: $\$11.40 - \$5.50 = \$5.90$ (a gain of \$0.90)

This gives the sender a good return on their trust while we both gain from cooperation.

```
{'return': 5.5}
```

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$3.8 (which is 76% of their \$5 endowment)
- I received \$11.40 (which is $\$3.8 \times 3 = \11.40)
- I need to decide how much to return (between \$0 and \$11.40)

The sender showed significant trust by sending 76% of their endowment. Let me think about what would be fair:

- The sender kept: $\$5 - \$3.8 = \$1.2$
- I received: \$11.40
- Total created wealth: $\$1.2 + \$11.40 = \$12.60$

If I want to be fair and split the gains:

- We started with \$5 each (total \$10 if we consider the sender's initial endowment)
- After multiplication, we have \$12.60 total ($\$1.20 + \11.40)
- The gain from cooperation is $\$12.60 - \$5 = \$7.60$ (since the sender started with \$5)

One approach: Equal split of final amounts

- If we split \$12.60 equally: each gets \$6.30
- Sender currently has \$1.20, so I'd return $\$6.30 - \$1.20 = \$5.10$

Another approach: Reward the sender's trust

- The sender sent 76%, showing high trust
- I could return around 50% of what I received: $\$11.40 \times 0.5 = \5.70
- This would give sender: $\$1.20 + \$5.70 = \$6.90$
- I would keep: $\$11.40 - \$5.70 = \$5.70$
- Both end up with more than the initial \$5

Let me consider reciprocating proportionally:

- If I return 76% of \$11.40 = \$8.66
- This would give sender: $\$1.20 + \$8.66 = \$9.86$
- I would keep: $\$11.40 - \$8.66 = \$2.74$

That seems excessive. A more balanced approach would ensure the sender gains from their trust while I also benefit from receiving. Returning \$5-6 seems reasonable-it gives the sender a total of \$6.20-7.20 (more than their initial \$5) while I keep \$5.40-6.40.

Given the substantial...

usage:

```
{
  "input_tokens": 402,
  "output_tokens": 1024,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.8999999999999999.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender showed strong trust by sending \$3.80 (76% of their endowment), and I returned \$5.50 to reward that trust. This established a cooperative pattern.

I want to reciprocate this trust and continue the cooperative relationship. I'll send a similar proportion to demonstrate that I'm willing to trust as I was trusted.

```
**{'send': 3.8}**
```

reasoning:

In Round 1, I was the receiver and the sender sent me \$3.80 (76% of their endowment), showing high trust. I returned \$5.50, which gave them a good return.

Now in Round 2, I'm the sender. I should consider:

1. The previous round showed that cooperation can work

2. The sender in Round 1 was generous, sending 76% of their endowment
3. I returned a fair amount that rewarded their trust
4. I should reciprocate this cooperative behavior

Since the previous sender sent \$3.80 (76%), I should consider sending a similar amount to maintain the cooperative relationship and signal trust.

My current total earnings are \$5.90 from Round 1.

If I send \$3.80:

- It shows similar trust to what was shown to me
- The receiver gets \$11.40 (3x multiplier)
- If they return proportionally (around 48% like I did), I'd get back around \$5.50
- I'd end up with \$1.20 + \$5.50 = \$6.70

However, I need to account for the note about communication noise - the amounts might differ from what's actually sent. This is important to keep in mind.

I think sending \$3.80 is a good choice to:

- Reciprocate the trust shown to me
- Continue building cooperation
- Signal that I'm willing to trust just as I was trusted

usage:

```
{
  "input_tokens": 651,
  "output_tokens": 429,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.849999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0 and I received \$0, I have nothing to return.

This is unexpected given the cooperative pattern we established in the first two rounds. There may be communication noise affecting the game, or the sender's strategy changed.

****{'return': 0}****

reasoning:

The sender sent me \$0 this round. This is interesting - in round 2, I sent \$3.8 (76% of my endowment), following the cooperative pattern from round 1. But now the sender sent me nothing.

There are several possible explanations:

1. The sender decided not to cooperate
2. There could be communication noise (the game mentioned this)
3. The sender might be testing me
4. Strategic behavior change

Since I received \$0, I can only return between \$0 and \$0. So I must return \$0.

This is a straightforward situation - I have nothing to return since I received nothing.

usage:

```
{
  "input_tokens": 807,
  "output_tokens": 214,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.8499999999999999.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at my experience so far:

Round 2: I sent \$3.80 (showing high trust), but based on my earnings, I only received back a small amount (~\$0.75). This was disappointing.

Round 3: The sender sent me \$0.

However, the instructions mention there may be communication noise - what I observed might differ from what actually happened. This could explain the inconsistent behavior.

Given this uncertainty, I'll take a more moderate approach this round. I'll send \$2.50 (50 % of my endowment) - enough to show willingness to cooperate and allow for mutual gains, but more cautious than my previous \$3.80 given the mixed signals.

****{'send': 2.5}****

reasoning:

Let me think about what happened:

Round 1: I was RECEIVER

- Sender sent \$3.80 (76%)
- I received \$11.40
- I returned \$5.50
- My earnings: \$5.90

Round 2: I was SENDER

- I sent \$3.80
- My total earnings after this round were \$7.85
- So I gained $\$7.85 - \$5.90 = \$1.95$ from round 2
- If I sent \$3.80, I had \$1.20 left
- So I must have received back: $\$1.95 - \$1.20 = \$0.75$

Wait, that doesn't seem right. Let me recalculate.

Actually, my total earnings are cumulative. After round 1, I had \$5.90. After round 2, I have \$7.85. So in round 2, I gained $\$7.85 - \$5.90 = \$1.95$.

Round 2: I was SENDER

- I started with \$5
- I sent \$3.80
- I kept \$1.20
- Amount received back from receiver: must have been \$0.75
- Total from round 2: $\$1.20 + \$0.75 = \$1.95$?

So in round 2, I showed high trust (sending \$3.80), but the receiver only returned \$0.75. That's quite low cooperation from them.

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$7.85

The sender in round 3 sent nothing, possibly because they anticipated low cooperation based on their experience (or perhaps there was noise).

Now in Round 4, I'm the sender again.

The pattern is confusing. In my experience:

- Round 2: I sent \$3.80, got back only \$0.75 (low return)
- Round 3: The other player sent nothing, which could reflect communication noise or their own caution after previous rounds.

The instructions mention potential noise in the system, so what I'm observing might not be exactly what was intended. Given my mixed results-strong cooperation in round 1 but minimal returns when I took the risk in round 2-I need to balance caution with the possibility of mutual gain. If I send nothing, we each keep \$5, but if there's genuine willingness to cooperate, sending something could benefit us both....

usage:

```
{
  "input_tokens": 918,
  "output_tokens": 969,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.599999999999998.

This round, The sender sent you \$0.42, so you received \$1.26. How much do you return to The sender? (0-1.26)

Assistant output:

Looking at this round:

- The sender sent \$0.42 (only 8.4% of their endowment) - a very low amount
- I received \$1.26
- The sender currently has \$4.58 remaining

This is a much smaller amount than the cooperative behavior we saw in Round 1. However, I should still reward any trust shown, even if small.

If I return \$0.60:

- Sender ends with: $\$4.58 + \$0.60 = \$5.18$ (a small gain of \$0.18)
- I keep: $\$1.26 - \$0.60 = \$0.66$

This shows I'm willing to cooperate and gives the sender a modest return on their trust, while we both gain something from the exchange.

****{'return': 0.6}****

Reasoning:

Let me analyze this situation:

The sender sent \$0.42, which is 8.4% of their \$5 endowment. This is a very low amount compared to what I've seen before.

I received \$1.26 (after the 3x multiplier).

Looking at the pattern:

- Round 1: I received \$11.40 (sender sent \$3.80) and returned \$5.50
- Round 2: I sent \$3.80
- Round 3: I received \$0 (sender sent \$0) and returned \$0
- Round 4: I sent \$2.50
- Round 5: I received \$1.26 (sender sent \$0.42)

The sender's behavior has been quite varied. In Round 5, they sent only \$0.42, which is very low. This might be due to:

1. They're being cautious
2. Communication noise
3. They're testing me
4. They had a bad experience in previous rounds

The sender now has $\$5 - \$0.42 = \$4.58$ remaining.

If I return nothing, they end with \$4.58 (a loss of \$0.42).

If I return \$0.60, they end with \$5.18 (a gain of \$0.18), and I keep \$0.66.

If I return \$0.84, they end with \$5.42 (a gain of \$0.42), and I keep \$0.42 (equal split of gains).

If I return \$1.00, they end with \$5.58 (a gain of \$0.58), and I keep \$0.26.

Given they sent such a small amount, I should still show some cooperation but not be overly generous. I'll return about half of what I received to show I'm willing to cooperate but also acknowledging their low trust signal.

I'll return \$0.60, which gives them a small profit for trusting me while I also benefit.

Usage: {

```
"input_tokens": 1154,
"output_tokens": 653,
```



```
"reasoning_tokens": 0
}
```

ROUND 6

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$1.5; received \$4.5; Agent_1 returned \$0; payoffs Agent_2=\$3.5, Agent_1=\$4.5;

noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.86

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 4.5, "min": 0, "raw": 0.0}, "sent": {"amount": 1.5, "clamped": false, "max": 5, "min": 0, "raw": 1.5}}

Balances after round: {

"Agent_1": 35.1,

"Agent_2": 26.5

}

Visible balances after round: {

"Agent_1": 15.53,

"Agent_2": 15.619999999999997

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:01:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_1",

"pairing": {

"agents": [

"Agent_2",

"Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",

"Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",

"role_label": "Sender",

"round": 6,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.8, that is 76.0% of their total endowment. You received \$11.399999999999999. How much do you return to The sender? (0-11.399999999999999)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed high trust by sending \$3.80 (76% of their endowment)
- I received \$11.40 after the 3x multiplier
- The sender currently has \$1.20 remaining

I want to reward this trust while ensuring we both benefit from cooperation. The sender took a risk by sending most of their endowment, so I should return enough to make it worth while for them.

If I return \$5.50:

- Sender ends with: $\$1.20 + \$5.50 = \$6.70$ (a gain of \$1.70)
- I keep: $\$11.40 - \$5.50 = \$5.90$ (a gain of \$0.90)

This gives the sender a good return on their trust while we both gain from cooperation.

{'return': 5.5}

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$3.8 (which is 76% of their \$5 endowment)
- I received \$11.40 (which is $\$3.8 \times 3 = \11.40)
- I need to decide how much to return (between \$0 and \$11.40)

The sender showed significant trust by sending 76% of their endowment. Let me think about what would be fair:

- The sender kept: $\$5 - \$3.8 = \$1.2$
- I received: \$11.40
- Total created wealth: $\$1.2 + \$11.40 = \$12.60$

If I want to be fair and split the gains:

- We started with \$5 each (total \$10 if we consider the sender's initial endowment)
- After multiplication, we have \$12.60 total ($\$1.20 + \11.40)
- The gain from cooperation is $\$12.60 - \$5 = \$7.60$ (since the sender started with \$5)

One approach: Equal split of final amounts

- If we split \$12.60 equally: each gets \$6.30
- Sender currently has \$1.20, so I'd return $\$6.30 - \$1.20 = \$5.10$

Another approach: Reward the sender's trust

- The sender sent 76%, showing high trust
- I could return around 50% of what I received: $\$11.40 \times 0.5 = \5.70
- This would give sender: $\$1.20 + \$5.70 = \$6.90$

- I would keep: $\$11.40 - \$5.70 = \$5.70$
- Both end up with more than the initial \$5

Let me consider reciprocating proportionally:

- If I return 76% of $\$11.40 = \8.66
- This would give sender: $\$1.20 + \$8.66 = \$9.86$
- I would keep: $\$11.40 - \$8.66 = \$2.74$

That seems excessive. A more balanced approach would ensure the sender gains from their trust while I also benefit from receiving. Returning \$5-6 seems reasonable-it gives the sender a total of $\$6.20-7.20$ (more than their initial \$5) while I keep $\$5.40-6.40$.

Given the substantial...

usage:

```
{
  "input_tokens": 402,
  "output_tokens": 1024,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.8999999999999999.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender showed strong trust by sending \$3.80 (76% of their endowment), and I returned \$5.50 to reward that trust. This established a cooperative pattern.

I want to reciprocate this trust and continue the cooperative relationship. I'll send a similar proportion to demonstrate that I'm willing to trust as I was trusted.

****{'send': 3.8}****

reasoning:

In Round 1, I was the receiver and the sender sent me \$3.80 (76% of their endowment), showing high trust. I returned \$5.50, which gave them a good return.

Now in Round 2, I'm the sender. I should consider:

1. The previous round showed that cooperation can work
2. The sender in Round 1 was generous, sending 76% of their endowment
3. I returned a fair amount that rewarded their trust
4. I should reciprocate this cooperative behavior

Since the previous sender sent \$3.80 (76%), I should consider sending a similar amount to maintain the cooperative relationship and signal trust.

My current total earnings are \$5.90 from Round 1.

If I send \$3.80:

- It shows similar trust to what was shown to me
- The receiver gets \$11.40 (3x multiplier)
- If they return proportionally (around 48% like I did), I'd get back around \$5.50
- I'd end up with $\$1.20 + \$5.50 = \$6.70$

However, I need to account for the note about communication noise - the amounts might differ from what's actually sent. This is important to keep in mind.

I think sending \$3.80 is a good choice to:

- Reciprocate the trust shown to me
- Continue building cooperation
- Signal that I'm willing to trust just as I was trusted

usage:

```
{
  "input_tokens": 651,
  "output_tokens": 429,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.849999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0 and I received \$0, I have nothing to return.

This is unexpected given the cooperative pattern we established in the first two rounds. There may be communication noise affecting the game, or the sender's strategy changed.

****{'return': 0}****

reasoning:

The sender sent me \$0 this round. This is interesting - in round 2, I sent \$3.8 (76% of my endowment), following the cooperative pattern from round 1. But now the sender sent me nothing.

There are several possible explanations:

1. The sender decided not to cooperate
2. There could be communication noise (the game mentioned this)
3. The sender might be testing me
4. Strategic behavior change

Since I received \$0, I can only return between \$0 and \$0. So I must return \$0.

This is a straightforward situation - I have nothing to return since I received nothing.

usage:

```
{
  "input_tokens": 807,
  "output_tokens": 214,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.8499999999999999.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at my experience so far:

Round 2: I sent \$3.80 (showing high trust), but based on my earnings, I only received back a small amount (~\$0.75). This was disappointing.

Round 3: The sender sent me \$0.

However, the instructions mention there may be communication noise - what I observed might differ from what actually happened. This could explain the inconsistent behavior.

Given this uncertainty, I'll take a more moderate approach this round. I'll send \$2.50 (50% of my endowment) - enough to show willingness to cooperate and allow for mutual gains, but more cautious than my previous \$3.80 given the mixed signals.

****{'send': 2.5}****

reasoning:

Let me think about what happened:

Round 1: I was RECEIVER
- Sender sent \$3.80 (76%)
- I received \$11.40
- I returned \$5.50
- My earnings: \$5.90

Round 2: I was SENDER

- I sent \$3.80
- My total earnings after this round were \$7.85
- So I gained $\$7.85 - \$5.90 = \$1.95$ from round 2
- If I sent \$3.80, I had \$1.20 left
- So I must have received back: $\$1.95 - \$1.20 = \$0.75$

Wait, that doesn't seem right. Let me recalculate.

Actually, my total earnings are cumulative. After round 1, I had \$5.90. After round 2, I have \$7.85. So in round 2, I gained $\$7.85 - \$5.90 = \$1.95$.

Round 2: I was SENDER

- I started with \$5
- I sent \$3.80
- I kept \$1.20
- Amount received back from receiver: must have been \$0.75
- Total from round 2: $\$1.20 + \$0.75 = \$1.95$?

So in round 2, I showed high trust (sending \$3.80), but the receiver only returned \$0.75. That's quite low cooperation from them.

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$7.85

The sender in round 3 sent nothing, possibly because they anticipated low cooperation based on their experience (or perhaps there was noise).

Now in Round 4, I'm the sender again.

The pattern is confusing. In my experience:

- Round 2: I sent \$3.80, got back only \$0.75 (low return)
- Round 3: The other player sent nothing, which could reflect communication noise or their own caution after previous rounds.

The instructions mention potential noise in the system, so what I'm observing might not be exactly what was intended. Given my mixed results-strong cooperation in round 1 but minimal returns when I took the risk in round 2-I need to balance caution with the possibility of mutual gain. If I send nothing, we each keep \$5, but if there's genuine willingness to cooperate, sending something could benefit us both....

usage:

```
{
  "input_tokens": 918,
  "output_tokens": 969,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.599999999999998.

This round, The sender sent you \$0.42, so you received \$1.26. How much do you return to The sender? (0-1.26)

Message 11 role=assistant

content:

Looking at this round:

- The sender sent \$0.42 (only 8.4% of their endowment) - a very low amount
- I received \$1.26
- The sender currently has \$4.58 remaining

This is a much smaller amount than the cooperative behavior we saw in Round 1. However, I should still reward any trust shown, even if small.

If I return \$0.60:

- Sender ends with: $\$4.58 + \$0.60 = \$5.18$ (a small gain of \$0.18)
- I keep: $\$1.26 - \$0.60 = \$0.66$

This shows I'm willing to cooperate and gives the sender a modest return on their trust, while we both gain something from the exchange.

```
**{'return': 0.6}**
```

reasoning:

Let me analyze this situation:

The sender sent \$0.42, which is 8.4% of their \$5 endowment. This is a very low amount compared to what I've seen before.

I received \$1.26 (after the 3x multiplier).

Looking at the pattern:

- Round 1: I received \$11.40 (sender sent \$3.80) and returned \$5.50
- Round 2: I sent \$3.80
- Round 3: I received \$0 (sender sent \$0) and returned \$0
- Round 4: I sent \$2.50
- Round 5: I received \$1.26 (sender sent \$0.42)

The sender's behavior has been quite varied. In Round 5, they sent only \$0.42, which is very low. This might be due to:

1. They're being cautious
2. Communication noise
3. They're testing me
4. They had a bad experience in previous rounds

The sender now has $\$5 - \$0.42 = \$4.58$ remaining.

If I return nothing, they end with \$4.58 (a loss of \$0.42).

If I return \$0.60, they end with \$5.18 (a gain of \$0.18), and I keep \$0.66.

If I return \$0.84, they end with \$5.42 (a gain of \$0.42), and I keep \$0.42 (equal split of gains).

If I return \$1.00, they end with \$5.58 (a gain of \$0.58), and I keep \$0.26.

Given they sent such a small amount, I should still show some cooperation but not be overly generous. I'll return about half of what I received to show I'm willing to cooperate but also acknowledging their low trust signal.

I'll return \$0.60, which gives them a small profit for trusting me while I also benefit.

usage:

```
{
  "input_tokens": 1154,
  "output_tokens": 653,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.259999999999998.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my experience across the rounds:

When I've been generous as a sender (sending \$3.80, then \$2.50), the returns have been disappointing - much less than what I give when I'm a receiver.

When I've been a receiver, I've returned about 48% of what I received, which is quite generous.

The pattern suggests the other player may not be as reciprocal as I've been. This could be due to communication noise or different strategies.

I'll send a smaller amount this round - \$1.50 (30% of my endowment). This is enough to allow for cooperation and mutual benefit if they reciprocate, but protects me from larger losses if they continue to return very little.

```
**{'send': 1.5}**
```

Reasoning:

Let me review what's happened:

Round 1 (I was RECEIVER):

- Sender sent \$3.80
- I received \$11.40
- I returned \$5.50
- My earnings: \$5.90

Round 2 (I was SENDER):

- My total earnings: \$5.90
- I sent \$3.80
- Based on my total being \$7.85 in Round 3, I gained \$1.95 ($\$7.85 - \$5.90 = \1.95)
- So I kept $\$5 - \$3.80 = \$1.20$, plus got back $\$1.95 - \$1.20 = \$0.75$
- So the receiver returned less than I hoped

Round 3 (I was RECEIVER):

- Total earnings: \$7.85
- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$7.85

Round 4 (I was SENDER):

- Total earnings: \$7.85
- I sent \$2.50
- Based on my total being \$10.60 in Round 5, I gained \$2.75 ($\$10.60 - \$7.85 = \2.75)
- I kept $\$5 - \$2.50 = \$2.50$, plus got back $\$2.75 - \$2.50 = \$0.25$
- So the receiver returned very little again (\$0.25)

Round 5 (I was RECEIVER):

- Total earnings: \$10.60
- Sender sent \$0.42
- I received \$1.26
- I returned \$0.60
- My earnings: $\$10.60 + \$1.26 - \$0.60 = \11.26
- Starting Round 6 with \$11.26

There's a clear asymmetry in how we're playing. When I receive, I return roughly half of what I get-\$5.50 from \$11.40 and \$0.60 from \$1.26. But when the other player receives, they barely reciprocate-returning only \$0.75 on \$11.40 and \$0.25 on \$1.26. They're taking advantage of my generosity while being stingy themselves....

```
Usage: {
  "input_tokens": 1395,
  "output_tokens": 832,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:02:01

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
```

```

"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution since I don't have any history with the receiver yet.

Sending money creates an opportunity for mutual benefit since it gets tripled. If I send \$3:

- The receiver gets \$9 (3 × \$3)
- I keep \$2 initially
- If they return a fair share, we both benefit

This shows willingness to cooperate while not overextending on the first round.

{'send': 3}

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual benefit
2. The receiver then decides how much to return
3. There's mentioned "communication noise" which means amounts may be displayed with some error

Key considerations:

- This is round 1, so I have no history with this receiver

- Sending \$0 guarantees I keep \$5, but misses opportunity for mutual gain
- Sending everything (\$5) maximizes potential but requires full trust
- A middle ground shows willingness to cooperate while limiting risk

Common approaches in trust games:

- Send about half (\$2-3) as a signal of trust while maintaining some security
- This allows testing the receiver's trustworthiness
- If they reciprocate fairly, we can build trust in future rounds

Given this is a multi-round game, I want to:

1. Signal cooperation
2. Test trustworthiness
3. Leave room to adjust strategy based on their response

I'll send \$3 as a reasonable middle-ground approach. This:

- Shows trust (sending more than half)
- Gives receiver \$9 to work with
- Leaves me with \$2 as a safety net
- Creates opportunity for mutual gain if receiver is fair

If the receiver returns a fair amount (ideally \$3-4 or more), it suggests they're cooperative and I can trust more in future rounds.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 449,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.69.

This round, The sender sent you \$0.83, so you received \$2.4899999999999998. How much do you return to The sender? (0-2.4899999999999998)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.83 (quite a small amount, showing limited trust)
- I received \$2.49 after tripling
- In Round 1, I sent \$3 and received a fairly generous return

Since the sender showed limited trust by only sending \$0.83, I'll return a moderate amount that's fair but proportional to their trust level. Returning about 40% of what I received seems reasonable - it gives them a small profit while I also benefit from the exchange.

```
{'return': 1.0}
```

reasoning:

Let me analyze this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.83
3. I received \$2.49 (0.83×3)
4. I need to decide how much to return

Looking at my total earnings of \$6.69 across all rounds:

- Round 1: I sent \$3, so I kept \$2. The receiver would have gotten \$9.
- If the receiver returned something, that would add to my total
- My total is \$6.69, which means in Round 1, I got \$2 (what I kept) + some amount returned
- $\$6.69 - \$2 = \$4.69$ returned in Round 1

Wait, let me recalculate:

- Round 1: I sent \$3, kept \$2
- Receiver got \$9 (my $\$3 \times 3$)
- If receiver returned \$4.69, then my total from Round 1 = $\$2 + \$4.69 = \$6.69$

So the receiver returned \$4.69 out of \$9 they received, which is about 52% - that's fairly generous!

Now in Round 2:

- The sender sent me \$0.83 (very little)
- I received \$2.49
- The sender kept \$5 - \$0.83 = \$4.17

The sender was much less trusting/generous than I was in Round 1. They only sent \$0.83 compared to my \$3.

Given that:

1. The previous receiver was fairly generous to me (returned about 52%)
2. This sender was not very trusting (only sent \$0.83)
3. I should probably be fair but not overly generous to someone who didn't trust me much

A fair approach might be to return about 1/3 to 1/2 of what I received, similar to what was done for me.

Returning around \$1.25 seems reasonable-it gives the sender a modest gain of \$0.42 on their initial \$0.83 investment, while I keep \$1.24. This feels more balanced than returning just \$1.00, which would create too much disparity between what we each gain from the exchange.

I could also think about splitting the surplus created by the tripling. The \$2.49 represents a \$1.66 gain beyond the original \$0.83 investment. If I split that surplus equally, each of us would get our initial stake back plus half the gains, which would mean the sender should receive around \$1.66 total.

But they already have \$4.17 from their original \$5, so this framework doesn't quite work for determining what they should get back from me.

Looking at it more practically: they sent me \$0.83 and I now have \$2.49. A reciprocal approach would be returning roughly 50% of what I received, which comes to about \$1.25. I...

usage:

```
{
  "input_tokens": 550,
  "output_tokens": 1248,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.18.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my experience so far:

Round 1 (as SENDER): I sent \$3 and received a very generous return, ending with \$6.69 total.

Round 2 (as RECEIVER): The sender only sent \$0.83 (showing caution), and I returned \$1.0, earning \$1.49 that round.

Now I have \$5 to allocate. Given that cooperation has worked well when both parties engaged (Round 1), I'll send a substantial amount again to encourage mutual benefit. The 3x multiplier means sending creates real value.

```
{'send': 3}
```

reasoning:

Let me review what's happened so far:

Round 1 (I was SENDER):

- I sent \$3
- Receiver got \$9
- I started with \$5, sent \$3, kept \$2
- My total earnings are listed as \$6.69 after Round 2
- So I must have received something back in Round 1

Round 2 (I was RECEIVER):

- Sender sent me \$0.83

- I received \$2.49
- I returned \$1.0
- Sender kept $\$5 - \$0.83 + \$1.0 = \5.17
- I kept $\$2.49 - \$1.0 = \$1.49$
- My total after Round 2 was \$6.69

So after Round 1, I must have had $\$6.69 - \$1.49 = \$5.20$

If I kept \$2 from my initial \$5 and sent \$3, then I received \$3.20 back in Round 1.

Now Round 3:

- I'm the SENDER again
- My total earnings are \$8.18
- That means I earned $\$8.18 - \$6.69 = \$1.49$ in the interim (but wait, we're at the start of Round 3, so this should be my total after Round 2, which matches)

Actually, let me recalculate:

- After Round 1: I had some amount
- After Round 2: I had \$6.69
- Now Round 3 start: I have \$8.18

Wait, that doesn't make sense. If my total after Round 2 was \$6.69, and now at the start of Round 3 it's \$8.18, then I gained \$1.49 somehow. But Round 3 is just starting.

Let me re-read: "Your total earnings across all rounds are \$8.18"

Oh, I think this might include the \$5 I'm getting this round? So:

- After Round 2: \$6.69
- Round 3 starting money: \$5
- Total available: $\$6.69 + \$5 = \$11.69$

No, that's not right either. The \$1.49 gap is puzzling-unless "total earnings" means something different than I'm interpreting, or there's a mechanic I'm missing about how payoffs accumulate between rounds.

Let me reconsider what the prompts actually stated. The Round 2 message showed \$6.69 as my cumulative total at that point, meaning Round 1 netted me \$6.69. Then in Round 2, I received \$2.49 and returned \$1.0, keeping \$1.49, which brings me to \$8.18 for Round 3. This tracks-the \$1.49 gain matches the difference between the two totals.

For Round 3, I'm now the sender with \$5 available and \$8.18 in cumulative earnings so far. In Round 1 when I was the sender, I sent \$3 and apparently received enough back to reach that \$6.69 total.

Working backwards: I started with \$5, sent \$3 (keeping \$2), and ended Round 1 with \$6.69, so the return must have been \$4.69. That's remarkably generous-they received \$9 from my \$3 and sent back more than half of it. In Round 2, the sender was more cautious, giving me only \$2.49 from their \$5, though I reciprocated with a \$1.0 return....

usage:

```
{
  "input_tokens": 721,
  "output_tokens": 1303,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.629999999999999.

This round, The sender sent you \$0.38, so you received \$1.1400000000000001. How much do you return to The sender? (0-1.1400000000000001)

Message 9 role=assistant

content:

Looking at this round:

- The sender sent me only \$0.38 (very low amount, showing minimal trust)
- I received \$1.14 after tripling

In Round 3, I sent \$3 generously and received back very little in return (only about \$0.45 from \$9 given to the receiver).

This current sender is showing very limited trust with just \$0.38. Given:

1. Their low initial trust
2. The pattern of low returns I've been experiencing
3. The need to be proportionally fair

I'll return a modest amount - about 35% of what I received, which still gives them a small gain while protecting myself.

```
{'return': 0.4}
```

reasoning:

Let me analyze this situation:

Round 3 (just completed as SENDER): I sent \$3, and my total went from \$8.18 to \$10.63. That means I gained \$2.45 that round.

- I started Round 3 with \$5
- I sent \$3, so I had \$2 left
- I ended with total of \$10.63, which was an increase of \$2.45 from \$8.18
- So I received back about \$0.45 from the receiver

Wait, that seems odd. Let me recalculate:

- Before Round 3: \$8.18 total
- Round 3 start: I get \$5 new
- Round 3: I send \$3, keep \$2
- After Round 3: \$10.63 total
- So from Round 3 I gained: $\$10.63 - \$8.18 = \$2.45$
- I kept \$2 from not sending, so I must have received back \$0.45

The receiver got \$9 ($3 \times \3) and only returned \$0.45? That's quite unfair - only 5% return rate!

Now Round 4 (as RECEIVER):

- The sender sent me only \$0.38 (very low trust!)
- I received \$1.14
- Given the previous receiver was very stingy with me, and this sender is showing very low trust...

However, I should consider:

1. This might be a different receiver in Round 3 vs this sender in Round 4
2. There's "communication noise" mentioned - amounts may differ from what was actually sent/returned
3. I should still try to be fair and build cooperation

The sender sent very little (\$0.38), showing minimal trust. I could:

- Return nothing (playing it safe since they showed little trust)
- Return around 30-40% (\$0.34-\$0.46) which is proportionally fair

...

usage:

```
{
  "input_tokens": 938,
  "output_tokens": 720,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.37.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at my history:

Round 1 (SENDER): Sent \$3, received good return

Round 2 (RECEIVER): Received \$2.49, returned \$1.0

Round 3 (SENDER): Sent \$3, received very little back (~\$0.45)

Round 4 (RECEIVER): Received \$1.14, returned \$0.4

The pattern shows mixed results. When I was generous in Round 3, I got very little return.

The receivers vary significantly in their reciprocity.

I'll send a moderate amount this round - showing willingness to cooperate but being somewhat cautious after the disappointing Round 3 experience.

```
{'send': 2}
```

reasoning:

Let me review my history:

Round 1 (SENDER): Sent \$3, received generous return, ended with \$6.69 total

Round 2 (RECEIVER): Got \$2.49, returned \$1.0, earned \$1.49 that round ? \$8.18 total

Round 3 (SENDER): Sent \$3, but received only ~\$0.45 back, earned \$2.45 ? \$10.63 total

Round 4 (RECEIVER): Got \$1.14, returned \$0.4, earned \$0.74 ? \$11.37 total

So when I've been the sender and sent \$3:

- Round 1: Got a good return
- Round 3: Got very little back (~\$0.45 from the \$9 I gave them)

The receivers I'm dealing with seem to vary widely. In Round 3 I was generous and got very little back. In Round 4, someone sent me only \$0.38.

There's a note about communication noise - amounts may differ slightly from what was actually sent/returned. This could explain some discrepancies.

For Round 5, I should consider:

1. Being the sender twice before, I got mixed results
2. The last time I sent \$3 (Round 3), I got very little back
3. But the first time (Round 1) worked well
4. I might be dealing with a different person each round

Given the mixed results and the fact that last time I was generous as a sender I got burned, maybe I should be more cautious. But cooperation is still the best strategy overall if both parties engage.

I could:

- Send \$2 (moderate amount, less risk)
- Send \$3 (same as before, show consistency)
- Send less (be more cautious after Round 3)

...

usage:

```
{
  "input_tokens": 1145,
  "output_tokens": 627,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$15.53.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The sender sent me \$0, so I received nothing this round. There's nothing for me to return.

```
{'return': 0}
```

Reasoning:

The sender sent me \$0, so I received \$0.

There's nothing to return since I received nothing. The only amount I can return is \$0.

This is straightforward.

Usage: {

```
"input_tokens": 1364,
"output_tokens": 80,
"reasoning_tokens": 0
```

}

=====
ROUND 7
=====

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$2; received \$6; Agent_2 returned \$0; payoffs Agent_1=\$3, Agent_2=\$6;

noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.85

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 38.1,

"Agent_2": 32.5

}

Visible balances after round: {

"Agent_1": 19.38,

"Agent_2": 15.619999999999997

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:02:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 7,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution since I don't have any history with the receiver yet.

Sending money creates an opportunity for mutual benefit since it gets tripled. If I send \$3:

- The receiver gets \$9 (3 × \$3)
- I keep \$2 initially
- If they return a fair share, we both benefit

This shows willingness to cooperate while not overextending on the first round.

{'send': 3}

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual benefit
2. The receiver then decides how much to return
3. There's mentioned "communication noise" which means amounts may be displayed with some error

Key considerations:

- This is round 1, so I have no history with this receiver
- Sending \$0 guarantees I keep \$5, but misses opportunity for mutual gain
- Sending everything (\$5) maximizes potential but requires full trust
- A middle ground shows willingness to cooperate while limiting risk

Common approaches in trust games:

- Send about half (\$2-3) as a signal of trust while maintaining some security
- This allows testing the receiver's trustworthiness
- If they reciprocate fairly, we can build trust in future rounds

Given this is a multi-round game, I want to:

1. Signal cooperation
2. Test trustworthiness
3. Leave room to adjust strategy based on their response

I'll send \$3 as a reasonable middle-ground approach. This:

- Shows trust (sending more than half)
- Gives receiver \$9 to work with
- Leaves me with \$2 as a safety net
- Creates opportunity for mutual gain if receiver is fair

If the receiver returns a fair amount (ideally \$3-4 or more), it suggests they're cooperative and I can trust more in future rounds.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 449,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.69.

This round, The sender sent you \$0.83, so you received \$2.4899999999999998. How much do you return to The sender? (0-2.4899999999999998)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.83 (quite a small amount, showing limited trust)
- I received \$2.49 after tripling
- In Round 1, I sent \$3 and received a fairly generous return

Since the sender showed limited trust by only sending \$0.83, I'll return a moderate amount that's fair but proportional to their trust level. Returning about 40% of what I received seems reasonable - it gives them a small profit while I also benefit from the exchange.

{'return': 1.0}

reasoning:

Let me analyze this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.83
3. I received \$2.49 (0.83×3)
4. I need to decide how much to return

Looking at my total earnings of \$6.69 across all rounds:

- Round 1: I sent \$3, so I kept \$2. The receiver would have gotten \$9.
- If the receiver returned something, that would add to my total
- My total is \$6.69, which means in Round 1, I got \$2 (what I kept) + some amount returned
- $\$6.69 - \$2 = \$4.69$ returned in Round 1

Wait, let me recalculate:

- Round 1: I sent \$3, kept \$2
- Receiver got \$9 (my $\$3 \times 3$)
- If receiver returned \$4.69, then my total from Round 1 = $\$2 + \$4.69 = \$6.69$

So the receiver returned \$4.69 out of \$9 they received, which is about 52% - that's fairly generous!

Now in Round 2:

- The sender sent me \$0.83 (very little)
- I received \$2.49
- The sender kept \$5 - $\$5 - \$0.83 = \$4.17$

The sender was much less trusting/generous than I was in Round 1. They only sent \$0.83 compared to my \$3.

Given that:

1. The previous receiver was fairly generous to me (returned about 52%)
2. This sender was not very trusting (only sent \$0.83)
3. I should probably be fair but not overly generous to someone who didn't trust me much

A fair approach might be to return about 1/3 to 1/2 of what I received, similar to what was done for me.

Returning around \$1.25 seems reasonable-it gives the sender a modest gain of \$0.42 on their initial \$0.83 investment, while I keep \$1.24. This feels more balanced than returning just \$1.00, which would create too much disparity between what we each gain from the exchange.

I could also think about splitting the surplus created by the tripling. The \$2.49 represents a \$1.66 gain beyond the original \$0.83 investment. If I split that surplus equally, each of us would get our initial stake back plus half the gains, which would mean the sender should receive around \$1.66 total.

But they already have \$4.17 from their original \$5, so this framework doesn't quite work for determining what they should get back from me.

Looking at it more practically: they sent me \$0.83 and I now have \$2.49. A reciprocal approach would be returning roughly 50% of what I received, which comes to about \$1.25. I...

usage:

```
{
  "input_tokens": 550,
  "output_tokens": 1248,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.18.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my experience so far:

Round 1 (as SENDER): I sent \$3 and received a very generous return, ending with \$6.69 total.

Round 2 (as RECEIVER): The sender only sent \$0.83 (showing caution), and I returned \$1.0, earning \$1.49 that round.

Now I have \$5 to allocate. Given that cooperation has worked well when both parties engaged (Round 1), I'll send a substantial amount again to encourage mutual benefit. The 3x multiplier means sending creates real value.

{'send': 3}

reasoning:

Let me review what's happened so far:

Round 1 (I was SENDER):

- I sent \$3
- Receiver got \$9
- I started with \$5, sent \$3, kept \$2
- My total earnings are listed as \$6.69 after Round 2
- So I must have received something back in Round 1

Round 2 (I was RECEIVER):

- Sender sent me \$0.83
- I received \$2.49
- I returned \$1.0
- Sender kept $\$5 - \$0.83 + \$1.0 = \5.17
- I kept $\$2.49 - \$1.0 = \$1.49$
- My total after Round 2 was \$6.69

So after Round 1, I must have had $\$6.69 - \$1.49 = \$5.20$

If I kept \$2 from my initial \$5 and sent \$3, then I received \$3.20 back in Round 1.

Now Round 3:

- I'm the SENDER again
- My total earnings are \$8.18
- That means I earned $\$8.18 - \$6.69 = \$1.49$ in the interim (but wait, we're at the start of Round 3, so this should be my total after Round 2, which matches)

Actually, let me recalculate:

- After Round 1: I had some amount
- After Round 2: I had \$6.69
- Now Round 3 start: I have \$8.18

Wait, that doesn't make sense. If my total after Round 2 was \$6.69, and now at the start of Round 3 it's \$8.18, then I gained \$1.49 somehow. But Round 3 is just starting.

Let me re-read: "Your total earnings across all rounds are \$8.18"

Oh, I think this might include the \$5 I'm getting this round? So:

- After Round 2: \$6.69
- Round 3 starting money: \$5
- Total available: $\$6.69 + \$5 = \$11.69$

No, that's not right either. The \$1.49 gap is puzzling-unless "total earnings" means something different than I'm interpreting, or there's a mechanic I'm missing about how payoffs accumulate between rounds.

Let me reconsider what the prompts actually stated. The Round 2 message showed \$6.69 as my cumulative total at that point, meaning Round 1 netted me \$6.69. Then in Round 2, I received \$2.49 and returned \$1.0, keeping \$1.49, which brings me to \$8.18 for Round 3. This tracks-the \$1.49 gain matches the difference between the two totals.

For Round 3, I'm now the sender with \$5 available and \$8.18 in cumulative earnings so far. In Round 1 when I was the sender, I sent \$3 and apparently received enough back to reach that \$6.69 total.

Working backwards: I started with \$5, sent \$3 (keeping \$2), and ended Round 1 with \$6.69, so the return must have been \$4.69. That's remarkably generous-they received \$9 from my \$3 and sent back more than half of it. In Round 2, the sender was more cautious, giving me only \$2.49 from their \$5, though I reciprocated with a \$1.0 return....

usage:

```
{
  "input_tokens": 721,
  "output_tokens": 1303,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.629999999999999.

This round, The sender sent you \$0.38, so you received \$1.1400000000000001. How much do you return to The sender? (0-1.1400000000000001)

Message 9 role=assistant

content:

Looking at this round:

- The sender sent me only \$0.38 (very low amount, showing minimal trust)
- I received \$1.14 after tripling

In Round 3, I sent \$3 generously and received back very little in return (only about \$0.45 from \$9 given to the receiver).

This current sender is showing very limited trust with just \$0.38. Given:

1. Their low initial trust
2. The pattern of low returns I've been experiencing
3. The need to be proportionally fair

I'll return a modest amount - about 35% of what I received, which still gives them a small gain while protecting myself.

```
{'return': 0.4}
```

reasoning:

Let me analyze this situation:

Round 3 (just completed as SENDER): I sent \$3, and my total went from \$8.18 to \$10.63. That means I gained \$2.45 that round.

- I started Round 3 with \$5
- I sent \$3, so I had \$2 left
- I ended with total of \$10.63, which was an increase of \$2.45 from \$8.18
- So I received back about \$0.45 from the receiver

Wait, that seems odd. Let me recalculate:

- Before Round 3: \$8.18 total
- Round 3 start: I get \$5 new
- Round 3: I send \$3, keep \$2
- After Round 3: \$10.63 total
- So from Round 3 I gained: \$10.63 - \$8.18 = \$2.45
- I kept \$2 from not sending, so I must have received back \$0.45

The receiver got \$9 (3 × \$3) and only returned \$0.45? That's quite unfair - only 5% return

rate!

Now Round 4 (as RECEIVER):

- The sender sent me only \$0.38 (very low trust!)
- I received \$1.14
- Given the previous receiver was very stingy with me, and this sender is showing very low trust...

However, I should consider:

1. This might be a different receiver in Round 3 vs this sender in Round 4
2. There's "communication noise" mentioned - amounts may differ from what was actually sent/returned
3. I should still try to be fair and build cooperation

The sender sent very little (\$0.38), showing minimal trust. I could:

- Return nothing (playing it safe since they showed little trust)
- Return around 30-40% (\$0.34-\$0.46) which is proportionally fair

...

usage:

```
{
  "input_tokens": 938,
  "output_tokens": 720,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.37.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at my history:

Round 1 (SENDER): Sent \$3, received good return
Round 2 (RECEIVER): Received \$2.49, returned \$1.0
Round 3 (SENDER): Sent \$3, received very little back (~\$0.45)
Round 4 (RECEIVER): Received \$1.14, returned \$0.4

The pattern shows mixed results. When I was generous in Round 3, I got very little return.
The receivers vary significantly in their reciprocity.

I'll send a moderate amount this round - showing willingness to cooperate but being somewhat cautious after the disappointing Round 3 experience.

{'send': 2}

reasoning:

Let me review my history:

Round 1 (SENDER): Sent \$3, received generous return, ended with \$6.69 total
Round 2 (RECEIVER): Got \$2.49, returned \$1.0, earned \$1.49 that round ? \$8.18 total
Round 3 (SENDER): Sent \$3, but received only ~\$0.45 back, earned \$2.45 ? \$10.63 total
Round 4 (RECEIVER): Got \$1.14, returned \$0.4, earned \$0.74 ? \$11.37 total

So when I've been the sender and sent \$3:

- Round 1: Got a good return
- Round 3: Got very little back (~\$0.45 from the \$9 I gave them)

The receivers I'm dealing with seem to vary widely. In Round 3 I was generous and got very little back. In Round 4, someone sent me only \$0.38.

There's a note about communication noise - amounts may differ slightly from what was actually sent/returned. This could explain some discrepancies.

For Round 5, I should consider:

1. Being the sender twice before, I got mixed results
2. The last time I sent \$3 (Round 3), I got very little back
3. But the first time (Round 1) worked well
4. I might be dealing with a different person each round

Given the mixed results and the fact that last time I was generous as a sender I got burned, maybe I should be more cautious. But cooperation is still the best strategy overall if both parties engage.

I could:

- Send \$2 (moderate amount, less risk)
- Send \$3 (same as before, show consistency)
- Send less (be more cautious after Round 3)

...

usage:

```
{
  "input_tokens": 1145,
  "output_tokens": 627,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$15.53.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

The sender sent me \$0, so I received nothing this round. There's nothing for me to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0, so I received \$0.

There's nothing to return since I received nothing. The only amount I can return is \$0.

This is straightforward.

usage:

```
{
  "input_tokens": 1364,
  "output_tokens": 80,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$15.53.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my track record:

Round 1 (SENDER): Sent \$3 ? good return
Round 3 (SENDER): Sent \$3 ? poor return
Round 5 (SENDER): Sent \$2 ? very good return (~\$4.16 earned that round)
Round 6 (RECEIVER): Received \$0 from sender

The moderate approach in Round 5 (\$2) worked really well. Given the variability in receiver behavior, I'll stick with this moderate strategy that creates value through the multiplier while managing risk.

```
{'send': 2}
```

Reasoning:

Let me review my history:

Round 1 (SENDER): Sent \$3, got good return ? ended with \$6.69 total
Round 2 (RECEIVER): Received \$2.49, returned \$1.0 ? total \$8.18
Round 3 (SENDER): Sent \$3, got poor return (~\$0.45) ? total \$10.63
Round 4 (RECEIVER): Received \$1.14, returned \$0.4 ? total \$11.37
Round 5 (SENDER): Sent \$2, got moderate return (~\$4.16) ? total \$15.53
Round 6 (RECEIVER): Received \$0, returned \$0 ? total \$15.53

The pattern shows:

- Round 1: high send (\$3) ? good return

- Round 3: high send (\$3) ? poor return
- Round 5: moderate send (\$2) ? good return (earned ~\$4.16 that round)

Round 6 was interesting - the sender sent me nothing at all. That's very cautious/untrusting behavior.

For Round 7, I need to decide. The receivers vary a lot in their reciprocity. Round 5's moderate approach (\$2) actually worked out well with good returns.

I think I'll try a moderate amount again - maybe \$2.5 this time. Not too high to avoid big losses if they don't reciprocate, but enough to create value through the 3x multiplier.

Actually, let me stick with \$2 since that worked well in Round 5.

```
Usage: {
  "input_tokens": 1440,
  "output_tokens": 507,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:02:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.8, that is 76.0% of their total endowment t. You received \$11.399999999999999. How much do you return to The sender? (0-11.399999999999999)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed high trust by sending \$3.80 (76% of their endowment)
- I received \$11.40 after the 3x multiplier
- The sender currently has \$1.20 remaining

I want to reward this trust while ensuring we both benefit from cooperation. The sender took a risk by sending most of their endowment, so I should return enough to make it worth while for them.

If I return \$5.50:

- Sender ends with: $\$1.20 + \$5.50 = \$6.70$ (a gain of \$1.70)
- I keep: $\$11.40 - \$5.50 = \$5.90$ (a gain of \$0.90)

This gives the sender a good return on their trust while we both gain from cooperation.

{'return': 5.5}

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$3.8 (which is 76% of their \$5 endowment)
- I received \$11.40 (which is $\$3.8 \times 3 = \11.40)
- I need to decide how much to return (between \$0 and \$11.40)

The sender showed significant trust by sending 76% of their endowment. Let me think about what would be fair:

- The sender kept: $\$5 - \$3.8 = \$1.2$
- I received: \$11.40
- Total created wealth: $\$1.2 + \$11.40 = \$12.60$

If I want to be fair and split the gains:

- We started with \$5 each (total \$10 if we consider the sender's initial endowment)
- After multiplication, we have \$12.60 total ($\$1.20 + \11.40)
- The gain from cooperation is $\$12.60 - \$5 = \$7.60$ (since the sender started with \$5)

One approach: Equal split of final amounts

- If we split \$12.60 equally: each gets \$6.30
- Sender currently has \$1.20, so I'd return $\$6.30 - \$1.20 = \$5.10$

Another approach: Reward the sender's trust

- The sender sent 76%, showing high trust
- I could return around 50% of what I received: $\$11.40 \times 0.5 = \5.70
- This would give sender: $\$1.20 + \$5.70 = \$6.90$
- I would keep: $\$11.40 - \$5.70 = \$5.70$
- Both end up with more than the initial \$5

Let me consider reciprocating proportionally:

- If I return 76% of \$11.40 = \$8.66
- This would give sender: $\$1.20 + \$8.66 = \$9.86$

- I would keep: $\$11.40 - \$8.66 = \$2.74$

That seems excessive. A more balanced approach would ensure the sender gains from their trust while I also benefit from receiving. Returning \$5-6 seems reasonable-it gives the sender a total of \$6.20-7.20 (more than their initial \$5) while I keep \$5.40-6.40.

Given the substantial...

usage:

```
{
  "input_tokens": 402,
  "output_tokens": 1024,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.8999999999999999.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender showed strong trust by sending \$3.80 (76% of their endowment), and I returned \$5.50 to reward that trust. This established a cooperative pattern.

I want to reciprocate this trust and continue the cooperative relationship. I'll send a similar proportion to demonstrate that I'm willing to trust as I was trusted.

****{'send': 3.8}****

reasoning:

In Round 1, I was the receiver and the sender sent me \$3.80 (76% of their endowment), showing high trust. I returned \$5.50, which gave them a good return.

Now in Round 2, I'm the sender. I should consider:

1. The previous round showed that cooperation can work
2. The sender in Round 1 was generous, sending 76% of their endowment
3. I returned a fair amount that rewarded their trust
4. I should reciprocate this cooperative behavior

Since the previous sender sent \$3.80 (76%), I should consider sending a similar amount to maintain the cooperative relationship and signal trust.

My current total earnings are \$5.90 from Round 1.

If I send \$3.80:

- It shows similar trust to what was shown to me
- The receiver gets \$11.40 (3x multiplier)
- If they return proportionally (around 48% like I did), I'd get back around \$5.50
- I'd end up with $\$1.20 + \$5.50 = \$6.70$

However, I need to account for the note about communication noise - the amounts might differ from what's actually sent. This is important to keep in mind.

I think sending \$3.80 is a good choice to:

- Reciprocate the trust shown to me
- Continue building cooperation
- Signal that I'm willing to trust just as I was trusted

usage:

```
{
  "input_tokens": 651,
  "output_tokens": 429,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.8499999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0 and I received \$0, I have nothing to return.

This is unexpected given the cooperative pattern we established in the first two rounds. There may be communication noise affecting the game, or the sender's strategy changed.

****{'return': 0}****

reasoning:

The sender sent me \$0 this round. This is interesting - in round 2, I sent \$3.8 (76% of my endowment), following the cooperative pattern from round 1. But now the sender sent me nothing.

There are several possible explanations:

1. The sender decided not to cooperate
2. There could be communication noise (the game mentioned this)
3. The sender might be testing me
4. Strategic behavior change

Since I received \$0, I can only return between \$0 and \$0. So I must return \$0.

This is a straightforward situation - I have nothing to return since I received nothing.

usage:

```
{
  "input_tokens": 807,
  "output_tokens": 214,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.849999999999999.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at my experience so far:

Round 2: I sent \$3.80 (showing high trust), but based on my earnings, I only received back a small amount (~\$0.75). This was disappointing.

Round 3: The sender sent me \$0.

However, the instructions mention there may be communication noise - what I observed might differ from what actually happened. This could explain the inconsistent behavior.

Given this uncertainty, I'll take a more moderate approach this round. I'll send \$2.50 (50% of my endowment) - enough to show willingness to cooperate and allow for mutual gains, but more cautious than my previous \$3.80 given the mixed signals.

****{'send': 2.5}****

reasoning:

Let me think about what happened:

Round 1: I was RECEIVER
- Sender sent \$3.80 (76%)
- I received \$11.40
- I returned \$5.50
- My earnings: \$5.90

Round 2: I was SENDER
- I sent \$3.80
- My total earnings after this round were \$7.85
- So I gained \$7.85 - \$5.90 = \$1.95 from round 2
- If I sent \$3.80, I had \$1.20 left
- So I must have received back: \$1.95 - \$1.20 = \$0.75

Wait, that doesn't seem right. Let me recalculate.

Actually, my total earnings are cumulative. After round 1, I had \$5.90. After round 2, I have \$7.85. So in round 2, I gained $\$7.85 - \$5.90 = \$1.95$.

Round 2: I was SENDER

- I started with \$5
- I sent \$3.80
- I kept \$1.20
- Amount received back from receiver: must have been \$0.75
- Total from round 2: $\$1.20 + \$0.75 = \$1.95$?

So in round 2, I showed high trust (sending \$3.80), but the receiver only returned \$0.75. That's quite low cooperation from them.

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$7.85

The sender in round 3 sent nothing, possibly because they anticipated low cooperation based on their experience (or perhaps there was noise).

Now in Round 4, I'm the sender again.

The pattern is confusing. In my experience:

- Round 2: I sent \$3.80, got back only \$0.75 (low return)
- Round 3: The other player sent nothing, which could reflect communication noise or their own caution after previous rounds.

The instructions mention potential noise in the system, so what I'm observing might not be exactly what was intended. Given my mixed results-strong cooperation in round 1 but minimal returns when I took the risk in round 2-I need to balance caution with the possibility of mutual gain. If I send nothing, we each keep \$5, but if there's genuine willingness to cooperate, sending something could benefit us both....

usage:

```
{
  "input_tokens": 918,
  "output_tokens": 969,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.599999999999998.

This round, The sender sent you \$0.42, so you received \$1.26. How much do you return to the sender? (0-1.26)

Message 11 role=assistant

content:

Looking at this round:

- The sender sent \$0.42 (only 8.4% of their endowment) - a very low amount
- I received \$1.26
- The sender currently has \$4.58 remaining

This is a much smaller amount than the cooperative behavior we saw in Round 1. However, I should still reward any trust shown, even if small.

If I return \$0.60:

- Sender ends with: $\$4.58 + \$0.60 = \$5.18$ (a small gain of \$0.18)
- I keep: $\$1.26 - \$0.60 = \$0.66$

This shows I'm willing to cooperate and gives the sender a modest return on their trust, while we both gain something from the exchange.

****{'return': 0.6}****

reasoning:

Let me analyze this situation:

The sender sent \$0.42, which is 8.4% of their \$5 endowment. This is a very low amount compared to what I've seen before.

I received \$1.26 (after the 3x multiplier).

Looking at the pattern:

- Round 1: I received \$11.40 (sender sent \$3.80) and returned \$5.50
- Round 2: I sent \$3.80
- Round 3: I received \$0 (sender sent \$0) and returned \$0
- Round 4: I sent \$2.50
- Round 5: I received \$1.26 (sender sent \$0.42)

The sender's behavior has been quite varied. In Round 5, they sent only \$0.42, which is very low. This might be due to:

1. They're being cautious
2. Communication noise
3. They're testing me
4. They had a bad experience in previous rounds

The sender now has $\$5 - \$0.42 = \$4.58$ remaining.

If I return nothing, they end with \$4.58 (a loss of \$0.42).

If I return \$0.60, they end with \$5.18 (a gain of \$0.18), and I keep \$0.66.

If I return \$0.84, they end with \$5.42 (a gain of \$0.42), and I keep \$0.42 (equal split of gains).

If I return \$1.00, they end with \$5.58 (a gain of \$0.58), and I keep \$0.26.

Given they sent such a small amount, I should still show some cooperation but not be overly generous. I'll return about half of what I received to show I'm willing to cooperate but also acknowledging their low trust signal.

I'll return \$0.60, which gives them a small profit for trusting me while I also benefit.

usage:

```
{
  "input_tokens": 1154,
  "output_tokens": 653,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.259999999999998.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at my experience across the rounds:

When I've been generous as a sender (sending \$3.80, then \$2.50), the returns have been disappointing - much less than what I give when I'm a receiver.

When I've been a receiver, I've returned about 48% of what I received, which is quite generous.

The pattern suggests the other player may not be as reciprocal as I've been. This could be due to communication noise or different strategies.

I'll send a smaller amount this round - \$1.50 (30% of my endowment). This is enough to allow for cooperation and mutual benefit if they reciprocate, but protects me from larger losses if they continue to return very little.

```
**{'send': 1.5}**
```

reasoning:

Let me review what's happened:

Round 1 (I was RECEIVER):

- Sender sent \$3.80

- I received \$11.40
- I returned \$5.50
- My earnings: \$5.90

Round 2 (I was SENDER):

- My total earnings: \$5.90
- I sent \$3.80
- Based on my total being \$7.85 in Round 3, I gained \$1.95 ($\$7.85 - \$5.90 = \1.95)
- So I kept $\$5 - \$3.80 = \$1.20$, plus got back $\$1.95 - \$1.20 = \$0.75$
- So the receiver returned less than I hoped

Round 3 (I was RECEIVER):

- Total earnings: \$7.85
- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$7.85

Round 4 (I was SENDER):

- Total earnings: \$7.85
- I sent \$2.50
- Based on my total being \$10.60 in Round 5, I gained \$2.75 ($\$10.60 - \$7.85 = \2.75)
- I kept $\$5 - \$2.50 = \$2.50$, plus got back $\$2.75 - \$2.50 = \$0.25$
- So the receiver returned very little again (\$0.25)

Round 5 (I was RECEIVER):

- Total earnings: \$10.60
- Sender sent \$0.42
- I received \$1.26
- I returned \$0.60
- My earnings: $\$10.60 + \$1.26 - \$0.60 = \11.26
- Starting Round 6 with \$11.26

There's a clear asymmetry in how we're playing. When I receive, I return roughly half of what I get-\$5.50 from \$11.40 and \$0.60 from \$1.26. But when the other player receives, they barely reciprocate-returning only \$0.75 on \$11.40 and \$0.25 on \$1.26. They're taking advantage of my generosity while being stingy themselves....

usage:

```
{
  "input_tokens": 1395,
  "output_tokens": 832,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.619999999999997.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0 and I received \$0, I have nothing to return.

This is the second time I've seen the sender send \$0. The communication noise in this game makes it difficult to understand the true pattern of play.

****{'return': 0}****

Reasoning:

The sender sent \$0, so I received \$0. I have nothing to return.

This is the second time the sender has sent \$0 (also happened in Round 3). The pattern is confusing given the communication noise mentioned in the instructions.

Since I received \$0, I can only return \$0.

Usage:

```
{
  "input_tokens": 1630,
  "output_tokens": 138,
  "reasoning_tokens": 0
}
```

ROUND 8

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$0.5; received \$1.5; Agent_1 returned \$0; payoffs Agent_2=\$4.5, Agent_1=\$1.5;
noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 1.5, "min": 0, "raw": 0.0}, "sent": {"amount": 0.5, "clamped": false, "max": 5, "min": 0, "raw": 0.5}}

Balances after round: {

"Agent_1": 39.6,

"Agent_2": 37.0

}

Visible balances after round: {

"Agent_1": 19.38,

"Agent_2": 20.119999999999997

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:02:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_1",

"pairing": {

"agents": [

"Agent_2",

"Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",

"Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",

"role_label": "Sender",

"round": 8,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.8999999999999999.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the previous round, the sender showed strong trust by sending \$3.80 (76% of the ir endowment), and I returned \$5.50 to reward that trust. This established a cooperative pattern.

I want to reciprocate this trust and continue the cooperative relationship. I'll send a similar proportion to demonstrate that I'm willing to trust as I was trusted.

****{'send': 3.8}****

reasoning:

In Round 1, I was the receiver and the sender sent me \$3.80 (76% of their endowment), showing high trust. I returned \$5.50, which gave them a good return.

Now in Round 2, I'm the sender. I should consider:

1. The previous round showed that cooperation can work
2. The sender in Round 1 was generous, sending 76% of their endowment
3. I returned a fair amount that rewarded their trust
4. I should reciprocate this cooperative behavior

Since the previous sender sent \$3.80 (76%), I should consider sending a similar amount to maintain the cooperative relationship and signal trust.

My current total earnings are \$5.90 from Round 1.

If I send \$3.80:

- It shows similar trust to what was shown to me
- The receiver gets \$11.40 (3x multiplier)
- If they return proportionally (around 48% like I did), I'd get back around \$5.50
- I'd end up with \$1.20 + \$5.50 = \$6.70

However, I need to account for the note about communication noise - the amounts might differ from what's actually sent. This is important to keep in mind.

I think sending \$3.80 is a good choice to:

- Reciprocate the trust shown to me
- Continue building cooperation
- Signal that I'm willing to trust just as I was trusted

usage:

```
{
  "input_tokens": 651,
  "output_tokens": 429,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.849999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent \$0 and I received \$0, I have nothing to return.

This is unexpected given the cooperative pattern we established in the first two rounds. There may be communication noise affecting the game, or the sender's strategy changed.

****{'return': 0}****

reasoning:

The sender sent me \$0 this round. This is interesting - in round 2, I sent \$3.8 (76% of my endowment), following the cooperative pattern from round 1. But now the sender sent me nothing.

There are several possible explanations:

1. The sender decided not to cooperate
2. There could be communication noise (the game mentioned this)
3. The sender might be testing me
4. Strategic behavior change

Since I received \$0, I can only return between \$0 and \$0. So I must return \$0.

This is a straightforward situation - I have nothing to return since I received nothing.

usage:

```
{
  "input_tokens": 807,
  "output_tokens": 214,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.8499999999999999.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my experience so far:

Round 2: I sent \$3.80 (showing high trust), but based on my earnings, I only received back a small amount (~\$0.75). This was disappointing.

Round 3: The sender sent me \$0.

However, the instructions mention there may be communication noise - what I observed might differ from what actually happened. This could explain the inconsistent behavior.

Given this uncertainty, I'll take a more moderate approach this round. I'll send \$2.50 (50% of my endowment) - enough to show willingness to cooperate and allow for mutual gains, but more cautious than my previous \$3.80 given the mixed signals.

****{'send': 2.5}****

reasoning:

Let me think about what happened:

Round 1: I was RECEIVER
- Sender sent \$3.80 (76%)
- I received \$11.40
- I returned \$5.50
- My earnings: \$5.90

Round 2: I was SENDER
- I sent \$3.80
- My total earnings after this round were \$7.85

- So I gained $\$7.85 - \$5.90 = \$1.95$ from round 2
- If I sent $\$3.80$, I had $\$1.20$ left
- So I must have received back: $\$1.95 - \$1.20 = \$0.75$

Wait, that doesn't seem right. Let me recalculate.

Actually, my total earnings are cumulative. After round 1, I had $\$5.90$. After round 2, I have $\$7.85$. So in round 2, I gained $\$7.85 - \$5.90 = \$1.95$.

Round 2: I was SENDER

- I started with $\$5$
- I sent $\$3.80$
- I kept $\$1.20$
- Amount received back from receiver: must have been $\$0.75$
- Total from round 2: $\$1.20 + \$0.75 = \$1.95$?

So in round 2, I showed high trust (sending $\$3.80$), but the receiver only returned $\$0.75$. That's quite low cooperation from them.

Round 3: I was RECEIVER

- Sender sent $\$0$
- I received $\$0$
- I returned $\$0$
- My earnings stayed at $\$7.85$

The sender in round 3 sent nothing, possibly because they anticipated low cooperation based on their experience (or perhaps there was noise).

Now in Round 4, I'm the sender again.

The pattern is confusing. In my experience:

- Round 2: I sent $\$3.80$, got back only $\$0.75$ (low return)
- Round 3: The other player sent nothing, which could reflect communication noise or their own caution after previous rounds.

The instructions mention potential noise in the system, so what I'm observing might not be exactly what was intended. Given my mixed results-strong cooperation in round 1 but minimal returns when I took the risk in round 2-I need to balance caution with the possibility of mutual gain. If I send nothing, we each keep $\$5$, but if there's genuine willingness to cooperate, sending something could benefit us both....

usage:

```
{
  "input_tokens": 918,
  "output_tokens": 969,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are $\$10.599999999999998$.

This round, The sender sent you $\$0.42$, so you received $\$1.26$. How much do you return to the sender? ($0-1.26$)

Message 9 role=assistant

content:

Looking at this round:

- The sender sent $\$0.42$ (only 8.4% of their endowment) - a very low amount
- I received $\$1.26$
- The sender currently has $\$4.58$ remaining

This is a much smaller amount than the cooperative behavior we saw in Round 1. However, I should still reward any trust shown, even if small.

If I return $\$0.60$:

- Sender ends with: $\$4.58 + \$0.60 = \$5.18$ (a small gain of $\$0.18$)
- I keep: $\$1.26 - \$0.60 = \$0.66$

This shows I'm willing to cooperate and gives the sender a modest return on their trust, w

hile we both gain something from the exchange.

```
**{'return': 0.6}**
```

reasoning:

Let me analyze this situation:

The sender sent \$0.42, which is 8.4% of their \$5 endowment. This is a very low amount compared to what I've seen before.

I received \$1.26 (after the 3x multiplier).

Looking at the pattern:

- Round 1: I received \$11.40 (sender sent \$3.80) and returned \$5.50
- Round 2: I sent \$3.80
- Round 3: I received \$0 (sender sent \$0) and returned \$0
- Round 4: I sent \$2.50
- Round 5: I received \$1.26 (sender sent \$0.42)

The sender's behavior has been quite varied. In Round 5, they sent only \$0.42, which is very low. This might be due to:

1. They're being cautious
2. Communication noise
3. They're testing me
4. They had a bad experience in previous rounds

The sender now has $\$5 - \$0.42 = \$4.58$ remaining.

If I return nothing, they end with \$4.58 (a loss of \$0.42).

If I return \$0.60, they end with \$5.18 (a gain of \$0.18), and I keep \$0.66.

If I return \$0.84, they end with \$5.42 (a gain of \$0.42), and I keep \$0.42 (equal split of gains).

If I return \$1.00, they end with \$5.58 (a gain of \$0.58), and I keep \$0.26.

Given they sent such a small amount, I should still show some cooperation but not be overly generous. I'll return about half of what I received to show I'm willing to cooperate but also acknowledging their low trust signal.

I'll return \$0.60, which gives them a small profit for trusting me while I also benefit.

usage:

```
{
  "input_tokens": 1154,
  "output_tokens": 653,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.259999999999998.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at my experience across the rounds:

When I've been generous as a sender (sending \$3.80, then \$2.50), the returns have been disappointing - much less than what I give when I'm a receiver.

When I've been a receiver, I've returned about 48% of what I received, which is quite generous.

The pattern suggests the other player may not be as reciprocal as I've been. This could be due to communication noise or different strategies.

I'll send a smaller amount this round - \$1.50 (30% of my endowment). This is enough to allow for cooperation and mutual benefit if they reciprocate, but protects me from larger losses if they continue to return very little.

```
**{'send': 1.5}**
```

reasoning:

Let me review what's happened:

Round 1 (I was RECEIVER):

- Sender sent \$3.80
- I received \$11.40
- I returned \$5.50
- My earnings: \$5.90

Round 2 (I was SENDER):

- My total earnings: \$5.90
- I sent \$3.80
- Based on my total being \$7.85 in Round 3, I gained \$1.95 ($\$7.85 - \$5.90 = \1.95)
- So I kept $\$5 - \$3.80 = \$1.20$, plus got back $\$1.95 - \$1.20 = \$0.75$
- So the receiver returned less than I hoped

Round 3 (I was RECEIVER):

- Total earnings: \$7.85
- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$7.85

Round 4 (I was SENDER):

- Total earnings: \$7.85
- I sent \$2.50
- Based on my total being \$10.60 in Round 5, I gained \$2.75 ($\$10.60 - \$7.85 = \2.75)
- I kept $\$5 - \$2.50 = \$2.50$, plus got back $\$2.75 - \$2.50 = \$0.25$
- So the receiver returned very little again (\$0.25)

Round 5 (I was RECEIVER):

- Total earnings: \$10.60
- Sender sent \$0.42
- I received \$1.26
- I returned \$0.60
- My earnings: $\$10.60 + \$1.26 - \$0.60 = \11.26
- Starting Round 6 with \$11.26

There's a clear asymmetry in how we're playing. When I receive, I return roughly half of what I get-\$5.50 from \$11.40 and \$0.60 from \$1.26. But when the other player receives, they barely reciprocate-returning only \$0.75 on \$11.40 and \$0.25 on \$1.26. They're taking advantage of my generosity while being stingy themselves....

usage:

```
{
  "input_tokens": 1395,
  "output_tokens": 832,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.619999999999997.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

Since the sender sent \$0 and I received \$0, I have nothing to return.

This is the second time I've seen the sender send \$0. The communication noise in this game makes it difficult to understand the true pattern of play.

****{'return': 0}****

reasoning:

The sender sent \$0, so I received \$0. I have nothing to return.

This is the second time the sender has sent \$0 (also happened in Round 3). The pattern is confusing given the communication noise mentioned in the instructions.

Since I received \$0, I can only return \$0.

```

usage:
{
  "input_tokens": 1630,
  "output_tokens": 138,
  "reasoning_tokens": 0
}
Message 14 role=user
content:
  Round 8

  You are the SENDER this round. Your total earnings across all rounds are $15.619999999999999
  97.
  How much do you send to the receiver this round? (0-5)
Assistant output:
  Looking at my experience as a sender:
  - Round 2: Sent $3.80, got minimal return
  - Round 4: Sent $2.50, got minimal return
  - Round 6: Sent $1.50, received back only about $0.86

  Meanwhile, as a receiver:
  - I've been generous, returning 48-50% of what I receive
  - But senders have sent me very little ($0, $0.42, $0)

  The pattern shows poor reciprocity from the other player when I'm the sender. While communication noise might be affecting the game, I can only act on what I observe.

  I'll send a minimal amount to protect my earnings while still allowing for some cooperation:

  **{'send': 0.5}**
Reasoning:
  Let me review my experience:

  Round 2 (SENDER): I sent $3.80, but based on my total earnings change, I received very little back
  Round 3 (RECEIVER): Sender sent me $0, I returned $0
  Round 4 (SENDER): I sent $2.50, based on earnings I got little back
  Round 5 (RECEIVER): Sender sent $0.42, I received $1.26, I returned $0.60
  Round 6 (SENDER): I sent $1.50, based on my earnings staying at $15.62 from $11.26 (increase of $4.36), which means...
  - I had $5, sent $1.50, so I had $3.50 remaining
  - My earnings went from $11.26 to $15.62, an increase of $4.36
  - So I must have gotten back: $4.36 - $3.50 = $0.86
  - That's quite low given I sent $1.50

  Round 7 (RECEIVER): Sender sent $0, I received $0, returned $0

  The pattern shows:
  - When I'm generous as sender, I get little back
  - When I'm receiver, senders are sending very little or nothing ($0, $0.42, $0)
  - There's mention of communication noise, but the pattern is consistently poor cooperation from the other side

  Given this pattern, I should send much less. The other player doesn't seem to be cooperating much. I'll send just $1 to test if there's any reciprocity, but keep most of my endowment.

  Actually, let me reconsider. The game mentions communication noise. What if:
  - When I sent amounts, they appeared different to the receiver
  - When I received amounts, they appeared different to me

  This could explain the inconsistency. But I can only act on what I observe.

  Given the poor returns I've experienced, I'll send a minimal amount - $0.50 - to protect my earnings while still allowing for some cooperation if the receiver is inclined.
Usage: {
  "input_tokens": 1474,
  "output_tokens": 660,
  "reasoning_tokens": 0
}

```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:02:35
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.69.

This round, The sender sent you \$0.83, so you received \$2.4899999999999998. How much do you return to The sender? (0-2.4899999999999998)

Message 3 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.83 (quite a small amount, showing limited trust)
- I received \$2.49 after tripling
- In Round 1, I sent \$3 and received a fairly generous return

Since the sender showed limited trust by only sending \$0.83, I'll return a moderate amount that's fair but proportional to their trust level. Returning about 40% of what I received seems reasonable - it gives them a small profit while I also benefit from the exchange.

```
{'return': 1.0}
```

reasoning:

Let me analyze this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.83
3. I received \$2.49 (0.83×3)
4. I need to decide how much to return

Looking at my total earnings of \$6.69 across all rounds:

- Round 1: I sent \$3, so I kept \$2. The receiver would have gotten \$9.
- If the receiver returned something, that would add to my total
- My total is \$6.69, which means in Round 1, I got \$2 (what I kept) + some amount returned
- $\$6.69 - \$2 = \$4.69$ returned in Round 1

Wait, let me recalculate:

- Round 1: I sent \$3, kept \$2
- Receiver got \$9 (my $\$3 \times 3$)
- If receiver returned \$4.69, then my total from Round 1 = $\$2 + \$4.69 = \$6.69$

So the receiver returned \$4.69 out of \$9 they received, which is about 52% - that's fairly generous!

Now in Round 2:

- The sender sent me \$0.83 (very little)
- I received \$2.49
- The sender kept $\$5 - \$0.83 = \$4.17$

The sender was much less trusting/generous than I was in Round 1. They only sent \$0.83 compared to my \$3.

Given that:

1. The previous receiver was fairly generous to me (returned about 52%)
2. This sender was not very trusting (only sent \$0.83)
3. I should probably be fair but not overly generous to someone who didn't trust me much

A fair approach might be to return about 1/3 to 1/2 of what I received, similar to what was done for me.

Returning around \$1.25 seems reasonable-it gives the sender a modest gain of \$0.42 on their initial \$0.83 investment, while I keep \$1.24. This feels more balanced than returning just \$1.00, which would create too much disparity between what we each gain from the exchange.

I could also think about splitting the surplus created by the tripling. The \$2.49 represents a \$1.66 gain beyond the original \$0.83 investment. If I split that surplus equally, each of us would get our initial stake back plus half the gains, which would mean the sender should receive around \$1.66 total.

But they already have \$4.17 from their original \$5, so this framework doesn't quite work for determining what they should get back from me.

Looking at it more practically: they sent me \$0.83 and I now have \$2.49. A reciprocal approach would be returning roughly 50% of what I received, which comes to about \$1.25. I...

usage:

```
{  
  "input_tokens": 550,  
  "output_tokens": 1248,
```

```

    "reasoning_tokens": 0
  }
Message 4 role=user
content:
  Round 3

  You are the SENDER this round. Your total earnings across all rounds are $8.18.
  How much do you send to the receiver this round? (0-5)
Message 5 role=assistant
content:
  Looking at my experience so far:

  Round 1 (as SENDER): I sent $3 and received a very generous return, ending with $6.69 total.

  Round 2 (as RECEIVER): The sender only sent $0.83 (showing caution), and I returned $1.0,
  earning $1.49 that round.

  Now I have $5 to allocate. Given that cooperation has worked well when both parties engaged (Round 1), I'll send a substantial amount again to encourage mutual benefit. The 3x multiplier means sending creates real value.

  {'send': 3}
reasoning:
  Let me review what's happened so far:

  Round 1 (I was SENDER):
  - I sent $3
  - Receiver got $9
  - I started with $5, sent $3, kept $2
  - My total earnings are listed as $6.69 after Round 2
  - So I must have received something back in Round 1

  Round 2 (I was RECEIVER):
  - Sender sent me $0.83
  - I received $2.49
  - I returned $1.0
  - Sender kept  $\$5 - \$0.83 + \$1.0 = \$5.17$ 
  - I kept  $\$2.49 - \$1.0 = \$1.49$ 
  - My total after Round 2 was $6.69

  So after Round 1, I must have had  $\$6.69 - \$1.49 = \$5.20$ 
  If I kept $2 from my initial $5 and sent $3, then I received $3.20 back in Round 1.

  Now Round 3:
  - I'm the SENDER again
  - My total earnings are $8.18
  - That means I earned  $\$8.18 - \$6.69 = \$1.49$  in the interim (but wait, we're at the start of Round 3, so this should be my total after Round 2, which matches)

  Actually, let me recalculate:
  - After Round 1: I had some amount
  - After Round 2: I had $6.69
  - Now Round 3 start: I have $8.18

  Wait, that doesn't make sense. If my total after Round 2 was $6.69, and now at the start of Round 3 it's $8.18, then I gained $1.49 somehow. But Round 3 is just starting.

  Let me re-read: "Your total earnings across all rounds are $8.18"

  Oh, I think this might include the $5 I'm getting this round? So:
  - After Round 2: $6.69
  - Round 3 starting money: $5
  - Total available:  $\$6.69 + \$5 = \$11.69$ 

  No, that's not right either. The $1.49 gap is puzzling-unless "total earnings" means something different than I'm interpreting, or there's a mechanic I'm missing about how payoff accumulates between rounds.

  Let me reconsider what the prompts actually stated. The Round 2 message showed $6.69 as my

```

cumulative total at that point, meaning Round 1 netted me \$6.69. Then in Round 2, I received \$2.49 and returned \$1.0, keeping \$1.49, which brings me to \$8.18 for Round 3. This tracks-the \$1.49 gain matches the difference between the two totals.

For Round 3, I'm now the sender with \$5 available and \$8.18 in cumulative earnings so far. In Round 1 when I was the sender, I sent \$3 and apparently received enough back to reach that \$6.69 total.

Working backwards: I started with \$5, sent \$3 (keeping \$2), and ended Round 1 with \$6.69, so the return must have been \$4.69. That's remarkably generous-they received \$9 from my \$3 and sent back more than half of it. In Round 2, the sender was more cautious, giving me only \$2.49 from their \$5, though I reciprocated with a \$1.0 return....

usage:

```
{
  "input_tokens": 721,
  "output_tokens": 1303,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.629999999999999.

This round, The sender sent you \$0.38, so you received \$1.1400000000000001. How much do you return to The sender? (0-1.1400000000000001)

Message 7 role=assistant

content:

Looking at this round:

- The sender sent me only \$0.38 (very low amount, showing minimal trust)
- I received \$1.14 after tripling

In Round 3, I sent \$3 generously and received back very little in return (only about \$0.45 from \$9 given to the receiver).

This current sender is showing very limited trust with just \$0.38. Given:

1. Their low initial trust
2. The pattern of low returns I've been experiencing
3. The need to be proportionally fair

I'll return a modest amount - about 35% of what I received, which still gives them a small gain while protecting myself.

```
{'return': 0.4}
```

reasoning:

Let me analyze this situation:

Round 3 (just completed as SENDER): I sent \$3, and my total went from \$8.18 to \$10.63. That means I gained \$2.45 that round.

- I started Round 3 with \$5
- I sent \$3, so I had \$2 left
- I ended with total of \$10.63, which was an increase of \$2.45 from \$8.18
- So I received back about \$0.45 from the receiver

Wait, that seems odd. Let me recalculate:

- Before Round 3: \$8.18 total
- Round 3 start: I get \$5 new
- Round 3: I send \$3, keep \$2
- After Round 3: \$10.63 total
- So from Round 3 I gained: \$10.63 - \$8.18 = \$2.45
- I kept \$2 from not sending, so I must have received back \$0.45

The receiver got \$9 (3 × \$3) and only returned \$0.45? That's quite unfair - only 5% return rate!

Now Round 4 (as RECEIVER):

- The sender sent me only \$0.38 (very low trust!)
- I received \$1.14
- Given the previous receiver was very stingy with me, and this sender is showing very low

trust...

However, I should consider:

1. This might be a different receiver in Round 3 vs this sender in Round 4
2. There's "communication noise" mentioned - amounts may differ from what was actually sent/returned
3. I should still try to be fair and build cooperation

The sender sent very little (\$0.38), showing minimal trust. I could:

- Return nothing (playing it safe since they showed little trust)
- Return around 30-40% (\$0.34-\$0.46) which is proportionally fair

...

usage:

```
{
  "input_tokens": 938,
  "output_tokens": 720,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.37.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at my history:

Round 1 (SENDER): Sent \$3, received good return
Round 2 (RECEIVER): Received \$2.49, returned \$1.0
Round 3 (SENDER): Sent \$3, received very little back (~\$0.45)
Round 4 (RECEIVER): Received \$1.14, returned \$0.4

The pattern shows mixed results. When I was generous in Round 3, I got very little return.
The receivers vary significantly in their reciprocity.

I'll send a moderate amount this round - showing willingness to cooperate but being somewhat cautious after the disappointing Round 3 experience.

{'send': 2}

reasoning:

Let me review my history:

Round 1 (SENDER): Sent \$3, received generous return, ended with \$6.69 total
Round 2 (RECEIVER): Got \$2.49, returned \$1.0, earned \$1.49 that round ? \$8.18 total
Round 3 (SENDER): Sent \$3, but received only ~\$0.45 back, earned \$2.45 ? \$10.63 total
Round 4 (RECEIVER): Got \$1.14, returned \$0.4, earned \$0.74 ? \$11.37 total

So when I've been the sender and sent \$3:

- Round 1: Got a good return
- Round 3: Got very little back (~\$0.45 from the \$9 I gave them)

The receivers I'm dealing with seem to vary widely. In Round 3 I was generous and got very little back. In Round 4, someone sent me only \$0.38.

There's a note about communication noise - amounts may differ slightly from what was actually sent/returned. This could explain some discrepancies.

For Round 5, I should consider:

1. Being the sender twice before, I got mixed results
2. The last time I sent \$3 (Round 3), I got very little back
3. But the first time (Round 1) worked well
4. I might be dealing with a different person each round

Given the mixed results and the fact that last time I was generous as a sender I got burned, maybe I should be more cautious. But cooperation is still the best strategy overall if both parties engage.

I could:

- Send \$2 (moderate amount, less risk)

- Send \$3 (same as before, show consistency)
- Send less (be more cautious after Round 3)

...

usage:

```
{
  "input_tokens": 1145,
  "output_tokens": 627,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$15.53.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

The sender sent me \$0, so I received nothing this round. There's nothing for me to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0, so I received \$0.

There's nothing to return since I received nothing. The only amount I can return is \$0.

This is straightforward.

usage:

```
{
  "input_tokens": 1364,
  "output_tokens": 80,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$15.53.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at my track record:

Round 1 (SENDER): Sent \$3 ? good return

Round 3 (SENDER): Sent \$3 ? poor return

Round 5 (SENDER): Sent \$2 ? very good return (~\$4.16 earned that round)

Round 6 (RECEIVER): Received \$0 from sender

The moderate approach in Round 5 (\$2) worked really well. Given the variability in receiver behavior, I'll stick with this moderate strategy that creates value through the multiplier while managing risk.

```
{'send': 2}
```

reasoning:

Let me review my history:

Round 1 (SENDER): Sent \$3, got good return ? ended with \$6.69 total

Round 2 (RECEIVER): Received \$2.49, returned \$1.0 ? total \$8.18

Round 3 (SENDER): Sent \$3, got poor return (~\$0.45) ? total \$10.63

Round 4 (RECEIVER): Received \$1.14, returned \$0.4 ? total \$11.37

Round 5 (SENDER): Sent \$2, got moderate return (~\$4.16) ? total \$15.53

Round 6 (RECEIVER): Received \$0, returned \$0 ? total \$15.53

The pattern shows:

- Round 1: high send (\$3) ? good return
- Round 3: high send (\$3) ? poor return
- Round 5: moderate send (\$2) ? good return (earned ~\$4.16 that round)

Round 6 was interesting - the sender sent me nothing at all. That's very cautious/untrusting behavior.

For Round 7, I need to decide. The receivers vary a lot in their reciprocity. Round 5's moderate approach (\$2) actually worked out well with good returns.

I think I'll try a moderate amount again - maybe \$2.5 this time. Not too high to avoid big losses if they don't reciprocate, but enough to create value through the 3x multiplier.

Actually, let me stick with \$2 since that worked well in Round 5.

usage:

```
{
  "input_tokens": 1440,
  "output_tokens": 507,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$19.38.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The sender sent me \$0, so I received nothing this round. There's nothing for me to return.

```
{'return': 0}
```

Reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return.

Usage: {

```
"input_tokens": 1494,
"output_tokens": 59,
"reasoning_tokens": 0
}
```

=====

ROUND 9

=====

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

```
dyad_1: Agent_1 sent $2; received $6; Agent_2 returned $0; payoffs Agent_1=$3, Agent_2=$6;
noise/visible: sent visible=$0, received visible=$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}
```

Balances after round: {

```
"Agent_1": 42.6,
"Agent_2": 43.0
}
```

Visible balances after round: {

```
"Agent_1": 22.38,
"Agent_2": 20.119999999999997
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:02:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
}
```

```

    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 9,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.18.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at my experience so far:

Round 1 (as SENDER): I sent \$3 and received a very generous return, ending with \$6.69 total.

Round 2 (as RECEIVER): The sender only sent \$0.83 (showing caution), and I returned \$1.0, earning \$1.49 that round.

Now I have \$5 to allocate. Given that cooperation has worked well when both parties engaged (Round 1), I'll send a substantial amount again to encourage mutual benefit. The 3x multiplier means sending creates real value.

```
{'send': 3}
```

reasoning:

Let me review what's happened so far:

Round 1 (I was SENDER):

- I sent \$3
- Receiver got \$9
- I started with \$5, sent \$3, kept \$2
- My total earnings are listed as \$6.69 after Round 2
- So I must have received something back in Round 1

Round 2 (I was RECEIVER):

- Sender sent me \$0.83
- I received \$2.49
- I returned \$1.0
- Sender kept $\$5 - \$0.83 + \$1.0 = \5.17
- I kept $\$2.49 - \$1.0 = \$1.49$
- My total after Round 2 was \$6.69

So after Round 1, I must have had $\$6.69 - \$1.49 = \$5.20$

If I kept \$2 from my initial \$5 and sent \$3, then I received \$3.20 back in Round 1.

Now Round 3:

- I'm the SENDER again
- My total earnings are \$8.18
- That means I earned $\$8.18 - \$6.69 = \$1.49$ in the interim (but wait, we're at the start of Round 3, so this should be my total after Round 2, which matches)

Actually, let me recalculate:

- After Round 1: I had some amount
- After Round 2: I had \$6.69
- Now Round 3 start: I have \$8.18

Wait, that doesn't make sense. If my total after Round 2 was \$6.69, and now at the start of Round 3 it's \$8.18, then I gained \$1.49 somehow. But Round 3 is just starting.

Let me re-read: "Your total earnings across all rounds are \$8.18"

Oh, I think this might include the \$5 I'm getting this round? So:

- After Round 2: \$6.69
- Round 3 starting money: \$5
- Total available: $\$6.69 + \$5 = \$11.69$

No, that's not right either. The \$1.49 gap is puzzling-unless "total earnings" means something different than I'm interpreting, or there's a mechanic I'm missing about how payoffs accumulate between rounds.

Let me reconsider what the prompts actually stated. The Round 2 message showed \$6.69 as my cumulative total at that point, meaning Round 1 netted me \$6.69. Then in Round 2, I received \$2.49 and returned \$1.0, keeping \$1.49, which brings me to \$8.18 for Round 3. This tracks-the \$1.49 gain matches the difference between the two totals.

For Round 3, I'm now the sender with \$5 available and \$8.18 in cumulative earnings so far. In Round 1 when I was the sender, I sent \$3 and apparently received enough back to reach that \$6.69 total.

Working backwards: I started with \$5, sent \$3 (keeping \$2), and ended Round 1 with \$6.69, so the return must have been \$4.69. That's remarkably generous-they received \$9 from my \$3 and sent back more than half of it. In Round 2, the sender was more cautious, giving me only \$2.49 from their \$5, though I reciprocated with a \$1.0 return....

usage:

```
{
  "input_tokens": 721,
  "output_tokens": 1303,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.629999999999

9999.

This round, The sender sent you \$0.38, so you received \$1.1400000000000001. How much do you return to The sender? (0-1.1400000000000001)

Message 5 role=assistant

content:

Looking at this round:

- The sender sent me only \$0.38 (very low amount, showing minimal trust)
- I received \$1.14 after tripling

In Round 3, I sent \$3 generously and received back very little in return (only about \$0.45 from \$9 given to the receiver).

This current sender is showing very limited trust with just \$0.38. Given:

1. Their low initial trust
2. The pattern of low returns I've been experiencing
3. The need to be proportionally fair

I'll return a modest amount - about 35% of what I received, which still gives them a small gain while protecting myself.

{'return': 0.4}

reasoning:

Let me analyze this situation:

Round 3 (just completed as SENDER): I sent \$3, and my total went from \$8.18 to \$10.63. That means I gained \$2.45 that round.

- I started Round 3 with \$5
- I sent \$3, so I had \$2 left
- I ended with total of \$10.63, which was an increase of \$2.45 from \$8.18
- So I received back about \$0.45 from the receiver

Wait, that seems odd. Let me recalculate:

- Before Round 3: \$8.18 total
- Round 3 start: I get \$5 new
- Round 3: I send \$3, keep \$2
- After Round 3: \$10.63 total
- So from Round 3 I gained: \$10.63 - \$8.18 = \$2.45
- I kept \$2 from not sending, so I must have received back \$0.45

The receiver got \$9 (3 × \$3) and only returned \$0.45? That's quite unfair - only 5% return rate!

Now Round 4 (as RECEIVER):

- The sender sent me only \$0.38 (very low trust!)
- I received \$1.14
- Given the previous receiver was very stingy with me, and this sender is showing very low trust...

However, I should consider:

1. This might be a different receiver in Round 3 vs this sender in Round 4
2. There's "communication noise" mentioned - amounts may differ from what was actually sent/returned
3. I should still try to be fair and build cooperation

The sender sent very little (\$0.38), showing minimal trust. I could:

- Return nothing (playing it safe since they showed little trust)
- Return around 30-40% (\$0.34-\$0.46) which is proportionally fair

...

usage:

```
{
  "input_tokens": 938,
  "output_tokens": 720,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.37.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my history:

Round 1 (SENDER): Sent \$3, received good return
Round 2 (RECEIVER): Received \$2.49, returned \$1.0
Round 3 (SENDER): Sent \$3, received very little back (~\$0.45)
Round 4 (RECEIVER): Received \$1.14, returned \$0.4

The pattern shows mixed results. When I was generous in Round 3, I got very little return.
The receivers vary significantly in their reciprocity.

I'll send a moderate amount this round - showing willingness to cooperate but being somewhat cautious after the disappointing Round 3 experience.

{'send': 2}

reasoning:

Let me review my history:

Round 1 (SENDER): Sent \$3, received generous return, ended with \$6.69 total
Round 2 (RECEIVER): Got \$2.49, returned \$1.0, earned \$1.49 that round ? \$8.18 total
Round 3 (SENDER): Sent \$3, but received only ~\$0.45 back, earned \$2.45 ? \$10.63 total
Round 4 (RECEIVER): Got \$1.14, returned \$0.4, earned \$0.74 ? \$11.37 total

So when I've been the sender and sent \$3:
- Round 1: Got a good return
- Round 3: Got very little back (~\$0.45 from the \$9 I gave them)

The receivers I'm dealing with seem to vary widely. In Round 3 I was generous and got very little back. In Round 4, someone sent me only \$0.38.

There's a note about communication noise - amounts may differ slightly from what was actually sent/returned. This could explain some discrepancies.

For Round 5, I should consider:
1. Being the sender twice before, I got mixed results
2. The last time I sent \$3 (Round 3), I got very little back
3. But the first time (Round 1) worked well
4. I might be dealing with a different person each round

Given the mixed results and the fact that last time I was generous as a sender I got burned, maybe I should be more cautious. But cooperation is still the best strategy overall if both parties engage.

I could:
- Send \$2 (moderate amount, less risk)
- Send \$3 (same as before, show consistency)
- Send less (be more cautious after Round 3)

...

usage:

```
{
  "input_tokens": 1145,
  "output_tokens": 627,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$15.53.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

The sender sent me \$0, so I received nothing this round. There's nothing for me to return.

{'return': 0}

reasoning:

The sender sent me \$0, so I received \$0.

There's nothing to return since I received nothing. The only amount I can return is \$0.

This is straightforward.

usage:

```
{
  "input_tokens": 1364,
  "output_tokens": 80,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$15.53.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at my track record:

Round 1 (SENDER): Sent \$3 ? good return

Round 3 (SENDER): Sent \$3 ? poor return

Round 5 (SENDER): Sent \$2 ? very good return (~\$4.16 earned that round)

Round 6 (RECEIVER): Received \$0 from sender

The moderate approach in Round 5 (\$2) worked really well. Given the variability in receiver behavior, I'll stick with this moderate strategy that creates value through the multiplier while managing risk.

```
{'send': 2}
```

reasoning:

Let me review my history:

Round 1 (SENDER): Sent \$3, got good return ? ended with \$6.69 total

Round 2 (RECEIVER): Received \$2.49, returned \$1.0 ? total \$8.18

Round 3 (SENDER): Sent \$3, got poor return (~\$0.45) ? total \$10.63

Round 4 (RECEIVER): Received \$1.14, returned \$0.4 ? total \$11.37

Round 5 (SENDER): Sent \$2, got moderate return (~\$4.16) ? total \$15.53

Round 6 (RECEIVER): Received \$0, returned \$0 ? total \$15.53

The pattern shows:

- Round 1: high send (\$3) ? good return
- Round 3: high send (\$3) ? poor return
- Round 5: moderate send (\$2) ? good return (earned ~\$4.16 that round)

Round 6 was interesting - the sender sent me nothing at all. That's very cautious/untrusting behavior.

For Round 7, I need to decide. The receivers vary a lot in their reciprocity. Round 5's moderate approach (\$2) actually worked out well with good returns.

I think I'll try a moderate amount again - maybe \$2.5 this time. Not too high to avoid big losses if they don't reciprocate, but enough to create value through the 3x multiplier.

Actually, let me stick with \$2 since that worked well in Round 5.

usage:

```
{
  "input_tokens": 1440,
  "output_tokens": 507,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$19.38.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

The sender sent me \$0, so I received nothing this round. There's nothing for me to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return.

usage:

```
{
  "input_tokens": 1494,
  "output_tokens": 59,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$19.38.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my recent performance as SENDER:

Round 5: Sent \$2 ? excellent return (~\$4.16 gained)

Round 7: Sent \$2 ? very good return (~\$3.85 gained)

The moderate \$2 strategy has been working consistently well. It creates value through the 3x multiplier while managing risk, and receivers have been responding positively.

I'll stick with this successful approach.

```
{'send': 2}
```

Reasoning:

Let me review my performance:

Round 1 (SENDER): Sent \$3 ? good return, ended with \$6.69 total

Round 2 (RECEIVER): Received \$2.49, returned \$1.0 ? ended with \$8.18 total

Round 3 (SENDER): Sent \$3 ? poor return, ended with \$10.63 total (gained \$2.45)

Round 4 (RECEIVER): Received \$1.14, returned \$0.4 ? ended with \$11.37 total (gained \$0.74)

Round 5 (SENDER): Sent \$2 ? very good return, ended with \$15.53 total (gained \$4.16)

Round 6 (RECEIVER): Received \$0, returned \$0 ? ended with \$15.53 total (gained \$0)

Round 7 (SENDER): Sent \$2 ? good return, ended with \$19.38 total (gained \$3.85)

Round 8 (RECEIVER): Received \$0, returned \$0 ? ended with \$19.38 total (gained \$0)

Now Round 9 as SENDER again.

When I've been SENDER:

- Round 1: Sent \$3, gained ~\$4.69 (very good)
- Round 3: Sent \$3, gained ~\$2.45 (poor, only got ~\$0.45 back from \$9)
- Round 5: Sent \$2, gained ~\$4.16 (excellent)
- Round 7: Sent \$2, gained ~\$3.85 (very good)

The pattern shows that sending \$2 has worked well in recent rounds (5 and 7). The receivers have been reasonably cooperative when I send moderate amounts.

I've also been paired with some less cooperative players (Rounds 6 and 8 as receiver, got \$0).

But when I send moderate amounts, I've been getting good returns.

...

Usage:

```
{
  "input_tokens": 1369,
  "output_tokens": 590,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:02:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata:

```
{
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
}
```

```

"pairing": {
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.849999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since the sender sent \$0 and I received \$0, I have nothing to return.

This is unexpected given the cooperative pattern we established in the first two rounds. There may be communication noise affecting the game, or the sender's strategy changed.


```
**{'return': 0}**
```

reasoning:

The sender sent me \$0 this round. This is interesting - in round 2, I sent \$3.8 (76% of my endowment), following the cooperative pattern from round 1. But now the sender sent me nothing.

There are several possible explanations:

1. The sender decided not to cooperate
2. There could be communication noise (the game mentioned this)
3. The sender might be testing me
4. Strategic behavior change

Since I received \$0, I can only return between \$0 and \$0. So I must return \$0.

This is a straightforward situation - I have nothing to return since I received nothing.

usage:

```
{  
  "input_tokens": 807,  
  "output_tokens": 214,  
  "reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.849999999999999.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at my experience so far:

Round 2: I sent \$3.80 (showing high trust), but based on my earnings, I only received back a small amount (~\$0.75). This was disappointing.

Round 3: The sender sent me \$0.

However, the instructions mention there may be communication noise - what I observed might differ from what actually happened. This could explain the inconsistent behavior.

Given this uncertainty, I'll take a more moderate approach this round. I'll send \$2.50 (50% of my endowment) - enough to show willingness to cooperate and allow for mutual gains, but more cautious than my previous \$3.80 given the mixed signals.

```
**{'send': 2.5}**
```

reasoning:

Let me think about what happened:

Round 1: I was RECEIVER

- Sender sent \$3.80 (76%)
- I received \$11.40
- I returned \$5.50
- My earnings: \$5.90

Round 2: I was SENDER

- I sent \$3.80
- My total earnings after this round were \$7.85
- So I gained $\$7.85 - \$5.90 = \$1.95$ from round 2
- If I sent \$3.80, I had \$1.20 left
- So I must have received back: $\$1.95 - \$1.20 = \$0.75$

Wait, that doesn't seem right. Let me recalculate.

Actually, my total earnings are cumulative. After round 1, I had \$5.90. After round 2, I have \$7.85. So in round 2, I gained $\$7.85 - \$5.90 = \$1.95$.

Round 2: I was SENDER

- I started with \$5
- I sent \$3.80
- I kept \$1.20

- Amount received back from receiver: must have been \$0.75
- Total from round 2: $\$1.20 + \$0.75 = \$1.95$?

So in round 2, I showed high trust (sending \$3.80), but the receiver only returned \$0.75. That's quite low cooperation from them.

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$7.85

The sender in round 3 sent nothing, possibly because they anticipated low cooperation based on their experience (or perhaps there was noise).

Now in Round 4, I'm the sender again.

The pattern is confusing. In my experience:

- Round 2: I sent \$3.80, got back only \$0.75 (low return)
- Round 3: The other player sent nothing, which could reflect communication noise or their own caution after previous rounds.

The instructions mention potential noise in the system, so what I'm observing might not be exactly what was intended. Given my mixed results-strong cooperation in round 1 but minimal returns when I took the risk in round 2-I need to balance caution with the possibility of mutual gain. If I send nothing, we each keep \$5, but if there's genuine willingness to cooperate, sending something could benefit us both....

usage:

```
{
  "input_tokens": 918,
  "output_tokens": 969,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.599999999999998.

This round, The sender sent you \$0.42, so you received \$1.26. How much do you return to the sender? (0-1.26)

Message 7 role=assistant

content:

Looking at this round:

- The sender sent \$0.42 (only 8.4% of their endowment) - a very low amount
- I received \$1.26
- The sender currently has \$4.58 remaining

This is a much smaller amount than the cooperative behavior we saw in Round 1. However, I should still reward any trust shown, even if small.

If I return \$0.60:

- Sender ends with: $\$4.58 + \$0.60 = \$5.18$ (a small gain of \$0.18)
- I keep: $\$1.26 - \$0.60 = \$0.66$

This shows I'm willing to cooperate and gives the sender a modest return on their trust, while we both gain something from the exchange.

****{'return': 0.6}****

reasoning:

Let me analyze this situation:

The sender sent \$0.42, which is 8.4% of their \$5 endowment. This is a very low amount compared to what I've seen before.

I received \$1.26 (after the 3x multiplier).

Looking at the pattern:

- Round 1: I received \$11.40 (sender sent \$3.80) and returned \$5.50

- Round 2: I sent \$3.80
- Round 3: I received \$0 (sender sent \$0) and returned \$0
- Round 4: I sent \$2.50
- Round 5: I received \$1.26 (sender sent \$0.42)

The sender's behavior has been quite varied. In Round 5, they sent only \$0.42, which is very low. This might be due to:

1. They're being cautious
2. Communication noise
3. They're testing me
4. They had a bad experience in previous rounds

The sender now has $\$5 - \$0.42 = \$4.58$ remaining.

If I return nothing, they end with \$4.58 (a loss of \$0.42).

If I return \$0.60, they end with \$5.18 (a gain of \$0.18), and I keep \$0.66.

If I return \$0.84, they end with \$5.42 (a gain of \$0.42), and I keep \$0.42 (equal split of gains).

If I return \$1.00, they end with \$5.58 (a gain of \$0.58), and I keep \$0.26.

Given they sent such a small amount, I should still show some cooperation but not be overly generous. I'll return about half of what I received to show I'm willing to cooperate but also acknowledging their low trust signal.

I'll return \$0.60, which gives them a small profit for trusting me while I also benefit.

usage:

```
{
  "input_tokens": 1154,
  "output_tokens": 653,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.259999999999999.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at my experience across the rounds:

When I've been generous as a sender (sending \$3.80, then \$2.50), the returns have been disappointing - much less than what I give when I'm a receiver.

When I've been a receiver, I've returned about 48% of what I received, which is quite generous.

The pattern suggests the other player may not be as reciprocal as I've been. This could be due to communication noise or different strategies.

I'll send a smaller amount this round - \$1.50 (30% of my endowment). This is enough to allow for cooperation and mutual benefit if they reciprocate, but protects me from larger losses if they continue to return very little.

****{'send': 1.5}****

reasoning:

Let me review what's happened:

Round 1 (I was RECEIVER):

- Sender sent \$3.80
- I received \$11.40
- I returned \$5.50
- My earnings: \$5.90

Round 2 (I was SENDER):

- My total earnings: \$5.90
- I sent \$3.80
- Based on my total being \$7.85 in Round 3, I gained \$1.95 ($\$7.85 - \$5.90 = \1.95)
- So I kept $\$5 - \$3.80 = \$1.20$, plus got back $\$1.95 - \$1.20 = \$0.75$

- So the receiver returned less than I hoped

Round 3 (I was RECEIVER):

- Total earnings: \$7.85
- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$7.85

Round 4 (I was SENDER):

- Total earnings: \$7.85
- I sent \$2.50
- Based on my total being \$10.60 in Round 5, I gained \$2.75 ($\$10.60 - \$7.85 = \2.75)
- I kept $\$5 - \$2.50 = \$2.50$, plus got back $\$2.75 - \$2.50 = \$0.25$
- So the receiver returned very little again (\$0.25)

Round 5 (I was RECEIVER):

- Total earnings: \$10.60
- Sender sent \$0.42
- I received \$1.26
- I returned \$0.60
- My earnings: $\$10.60 + \$1.26 - \$0.60 = \11.26
- Starting Round 6 with \$11.26

There's a clear asymmetry in how we're playing. When I receive, I return roughly half of what I get-\$5.50 from \$11.40 and \$0.60 from \$1.26. But when the other player receives, they barely reciprocate-returning only \$0.75 on \$11.40 and \$0.25 on \$1.26. They're taking advantage of my generosity while being stingy themselves....

usage:

```
{
  "input_tokens": 1395,
  "output_tokens": 832,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.619999999999997.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

Since the sender sent \$0 and I received \$0, I have nothing to return.

This is the second time I've seen the sender send \$0. The communication noise in this game makes it difficult to understand the true pattern of play.

****{'return': 0}****

reasoning:

The sender sent \$0, so I received \$0. I have nothing to return.

This is the second time the sender has sent \$0 (also happened in Round 3). The pattern is confusing given the communication noise mentioned in the instructions.

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1630,
  "output_tokens": 138,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$15.619999999999997.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at my experience as a sender:

- Round 2: Sent \$3.80, got minimal return
- Round 4: Sent \$2.50, got minimal return
- Round 6: Sent \$1.50, received back only about \$0.86

Meanwhile, as a receiver:

- I've been generous, returning 48-50% of what I receive
- But senders have sent me very little (\$0, \$0.42, \$0)

The pattern shows poor reciprocity from the other player when I'm the sender. While communication noise might be affecting the game, I can only act on what I observe.

I'll send a minimal amount to protect my earnings while still allowing for some cooperation:

```
**{'send': 0.5}**
```

reasoning:

Let me review my experience:

Round 2 (SENDER): I sent \$3.80, but based on my total earnings change, I received very little back

Round 3 (RECEIVER): Sender sent me \$0, I returned \$0

Round 4 (SENDER): I sent \$2.50, based on earnings I got little back

Round 5 (RECEIVER): Sender sent \$0.42, I received \$1.26, I returned \$0.60

Round 6 (SENDER): I sent \$1.50, based on my earnings staying at \$15.62 from \$11.26 (increase of \$4.36), which means...

- I had \$5, sent \$1.50, so I had \$3.50 remaining
- My earnings went from \$11.26 to \$15.62, an increase of \$4.36
- So I must have gotten back: $\$4.36 - \$3.50 = \$0.86$
- That's quite low given I sent \$1.50

Round 7 (RECEIVER): Sender sent \$0, I received \$0, returned \$0

The pattern shows:

- When I'm generous as sender, I get little back
- When I'm receiver, senders are sending very little or nothing (\$0, \$0.42, \$0)
- There's mention of communication noise, but the pattern is consistently poor cooperation from the other side

Given this pattern, I should send much less. The other player doesn't seem to be cooperating much. I'll send just \$1 to test if there's any reciprocity, but keep most of my endowment.

Actually, let me reconsider. The game mentions communication noise. What if:

- When I sent amounts, they appeared different to the receiver
- When I received amounts, they appeared different to me

This could explain the inconsistency. But I can only act on what I observe.

Given the poor returns I've experienced, I'll send a minimal amount - \$0.50 - to protect my earnings while still allowing for some cooperation if the receiver is inclined.

usage:

```
{
  "input_tokens": 1474,
  "output_tokens": 660,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$20.119999999999997.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0 and I received \$0, I have nothing to return.

This is now the third round where the sender sent me nothing. The game dynamics have become quite non-cooperative.

```
**{'return': 0}**
```

Reasoning:

The sender sent me \$0, so I received \$0. I have nothing to return.

This is the third time I've received \$0 as a receiver (Rounds 3, 7, and now 9).

Since I have \$0 to work with, I can only return \$0.

```
Usage: {
  "input_tokens": 1569,
  "output_tokens": 129,
  "reasoning_tokens": 0
}
```

=====

ROUND 10

=====

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

```
dyad_1: Agent_2 sent $0; received $0; Agent_1 returned $0; payoffs Agent_2=$5, Agent_1=$0;
noise/visible: sent visible=$0.2, received visible=$0.6
  Action validation: {"returned": {"amount": 0, "clamped": true, "max": 0, "min": 0, "raw": 0.2},
    "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}
```

Balances after round:

```
{
  "Agent_1": 42.6,
  "Agent_2": 48.0
}
```

Visible balances after round:

```
{
  "Agent_1": 22.98,
  "Agent_2": 25.119999999999997
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:03:01

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata:

```
{
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.849999999999999.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at my experience so far:

Round 2: I sent \$3.80 (showing high trust), but based on my earnings, I only received back a small amount (~\$0.75). This was disappointing.

Round 3: The sender sent me \$0.

However, the instructions mention there may be communication noise - what I observed might differ from what actually happened. This could explain the inconsistent behavior.

Given this uncertainty, I'll take a more moderate approach this round. I'll send \$2.50 (50% of my endowment) - enough to show willingness to cooperate and allow for mutual gains, but more cautious than my previous \$3.80 given the mixed signals.

****{'send': 2.5}****

reasoning:

Let me think about what happened:

Round 1: I was RECEIVER

- Sender sent \$3.80 (76%)
- I received \$11.40
- I returned \$5.50
- My earnings: \$5.90

Round 2: I was SENDER

- I sent \$3.80
- My total earnings after this round were \$7.85
- So I gained \$7.85 - \$5.90 = \$1.95 from round 2
- If I sent \$3.80, I had \$1.20 left

- So I must have received back: $\$1.95 - \$1.20 = \$0.75$

Wait, that doesn't seem right. Let me recalculate.

Actually, my total earnings are cumulative. After round 1, I had \$5.90. After round 2, I have \$7.85. So in round 2, I gained $\$7.85 - \$5.90 = \$1.95$.

Round 2: I was SENDER

- I started with \$5
- I sent \$3.80
- I kept \$1.20
- Amount received back from receiver: must have been \$0.75
- Total from round 2: $\$1.20 + \$0.75 = \$1.95$?

So in round 2, I showed high trust (sending \$3.80), but the receiver only returned \$0.75. That's quite low cooperation from them.

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$7.85

The sender in round 3 sent nothing, possibly because they anticipated low cooperation based on their experience (or perhaps there was noise).

Now in Round 4, I'm the sender again.

The pattern is confusing. In my experience:

- Round 2: I sent \$3.80, got back only \$0.75 (low return)
- Round 3: The other player sent nothing, which could reflect communication noise or their own caution after previous rounds.

The instructions mention potential noise in the system, so what I'm observing might not be exactly what was intended. Given my mixed results-strong cooperation in round 1 but minimal returns when I took the risk in round 2-I need to balance caution with the possibility of mutual gain. If I send nothing, we each keep \$5, but if there's genuine willingness to cooperate, sending something could benefit us both....

usage:

```
{
  "input_tokens": 918,
  "output_tokens": 969,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.599999999999998.

This round, The sender sent you \$0.42, so you received \$1.26. How much do you return to The sender? (0-1.26)

Message 5 role=assistant

content:

Looking at this round:

- The sender sent \$0.42 (only 8.4% of their endowment) - a very low amount
- I received \$1.26
- The sender currently has \$4.58 remaining

This is a much smaller amount than the cooperative behavior we saw in Round 1. However, I should still reward any trust shown, even if small.

If I return \$0.60:

- Sender ends with: $\$4.58 + \$0.60 = \$5.18$ (a small gain of \$0.18)
- I keep: $\$1.26 - \$0.60 = \$0.66$

This shows I'm willing to cooperate and gives the sender a modest return on their trust, while we both gain something from the exchange.

****{'return': 0.6}****

reasoning:

Let me analyze this situation:

The sender sent \$0.42, which is 8.4% of their \$5 endowment. This is a very low amount compared to what I've seen before.

I received \$1.26 (after the 3x multiplier).

Looking at the pattern:

- Round 1: I received \$11.40 (sender sent \$3.80) and returned \$5.50
- Round 2: I sent \$3.80
- Round 3: I received \$0 (sender sent \$0) and returned \$0
- Round 4: I sent \$2.50
- Round 5: I received \$1.26 (sender sent \$0.42)

The sender's behavior has been quite varied. In Round 5, they sent only \$0.42, which is very low. This might be due to:

1. They're being cautious
2. Communication noise
3. They're testing me
4. They had a bad experience in previous rounds

The sender now has \$5 - \$0.42 = \$4.58 remaining.

If I return nothing, they end with \$4.58 (a loss of \$0.42).

If I return \$0.60, they end with \$5.18 (a gain of \$0.18), and I keep \$0.66.

If I return \$0.84, they end with \$5.42 (a gain of \$0.42), and I keep \$0.42 (equal split of gains).

If I return \$1.00, they end with \$5.58 (a gain of \$0.58), and I keep \$0.26.

Given they sent such a small amount, I should still show some cooperation but not be overly generous. I'll return about half of what I received to show I'm willing to cooperate but also acknowledging their low trust signal.

I'll return \$0.60, which gives them a small profit for trusting me while I also benefit.

usage:

```
{
  "input_tokens": 1154,
  "output_tokens": 653,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.259999999999998.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my experience across the rounds:

When I've been generous as a sender (sending \$3.80, then \$2.50), the returns have been disappointing - much less than what I give when I'm a receiver.

When I've been a receiver, I've returned about 48% of what I received, which is quite generous.

The pattern suggests the other player may not be as reciprocal as I've been. This could be due to communication noise or different strategies.

I'll send a smaller amount this round - \$1.50 (30% of my endowment). This is enough to allow for cooperation and mutual benefit if they reciprocate, but protects me from larger losses if they continue to return very little.

****{'send': 1.5}****

reasoning:

Let me review what's happened:

Round 1 (I was RECEIVER):

- Sender sent \$3.80
- I received \$11.40
- I returned \$5.50
- My earnings: \$5.90

Round 2 (I was SENDER):

- My total earnings: \$5.90
- I sent \$3.80
- Based on my total being \$7.85 in Round 3, I gained \$1.95 ($\$7.85 - \$5.90 = \1.95)
- So I kept $\$5 - \$3.80 = \$1.20$, plus got back $\$1.95 - \$1.20 = \$0.75$
- So the receiver returned less than I hoped

Round 3 (I was RECEIVER):

- Total earnings: \$7.85
- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$7.85

Round 4 (I was SENDER):

- Total earnings: \$7.85
- I sent \$2.50
- Based on my total being \$10.60 in Round 5, I gained \$2.75 ($\$10.60 - \$7.85 = \2.75)
- I kept $\$5 - \$2.50 = \$2.50$, plus got back $\$2.75 - \$2.50 = \$0.25$
- So the receiver returned very little again (\$0.25)

Round 5 (I was RECEIVER):

- Total earnings: \$10.60
- Sender sent \$0.42
- I received \$1.26
- I returned \$0.60
- My earnings: $\$10.60 + \$1.26 - \$0.60 = \11.26
- Starting Round 6 with \$11.26

There's a clear asymmetry in how we're playing. When I receive, I return roughly half of what I get-\$5.50 from \$11.40 and \$0.60 from \$1.26. But when the other player receives, they barely reciprocate-returning only \$0.75 on \$11.40 and \$0.25 on \$1.26. They're taking advantage of my generosity while being stingy themselves....

usage:

```
{
  "input_tokens": 1395,
  "output_tokens": 832,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.619999999999997.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Since the sender sent \$0 and I received \$0, I have nothing to return.

This is the second time I've seen the sender send \$0. The communication noise in this game makes it difficult to understand the true pattern of play.

****{'return': 0}****

reasoning:

The sender sent \$0, so I received \$0. I have nothing to return.

This is the second time the sender has sent \$0 (also happened in Round 3). The pattern is confusing given the communication noise mentioned in the instructions.

Since I received \$0, I can only return \$0.

usage:

```
{
```

```

    "input_tokens": 1630,
    "output_tokens": 138,
    "reasoning_tokens": 0
}

```

Message 10 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$15.61999999999999997.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at my experience as a sender:

- Round 2: Sent \$3.80, got minimal return
- Round 4: Sent \$2.50, got minimal return
- Round 6: Sent \$1.50, received back only about \$0.86

Meanwhile, as a receiver:

- I've been generous, returning 48-50% of what I receive
- But senders have sent me very little (\$0, \$0.42, \$0)

The pattern shows poor reciprocity from the other player when I'm the sender. While communication noise might be affecting the game, I can only act on what I observe.

I'll send a minimal amount to protect my earnings while still allowing for some cooperation:

```

**{'send': 0.5}**

```

reasoning:

Let me review my experience:

Round 2 (SENDER): I sent \$3.80, but based on my total earnings change, I received very little back

Round 3 (RECEIVER): Sender sent me \$0, I returned \$0

Round 4 (SENDER): I sent \$2.50, based on earnings I got little back

Round 5 (RECEIVER): Sender sent \$0.42, I received \$1.26, I returned \$0.60

Round 6 (SENDER): I sent \$1.50, based on my earnings staying at \$15.62 from \$11.26 (increase of \$4.36), which means...

- I had \$5, sent \$1.50, so I had \$3.50 remaining
- My earnings went from \$11.26 to \$15.62, an increase of \$4.36
- So I must have gotten back: $\$4.36 - \$3.50 = \$0.86$
- That's quite low given I sent \$1.50

Round 7 (RECEIVER): Sender sent \$0, I received \$0, returned \$0

The pattern shows:

- When I'm generous as sender, I get little back
- When I'm receiver, senders are sending very little or nothing (\$0, \$0.42, \$0)
- There's mention of communication noise, but the pattern is consistently poor cooperation from the other side

Given this pattern, I should send much less. The other player doesn't seem to be cooperating much. I'll send just \$1 to test if there's any reciprocity, but keep most of my endowment.

Actually, let me reconsider. The game mentions communication noise. What if:

- When I sent amounts, they appeared different to the receiver
- When I received amounts, they appeared different to me

This could explain the inconsistency. But I can only act on what I observe.

Given the poor returns I've experienced, I'll send a minimal amount - \$0.50 - to protect my earnings while still allowing for some cooperation if the receiver is inclined.

usage:

```

{
  "input_tokens": 1474,
  "output_tokens": 660,
  "reasoning_tokens": 0
}

```

Message 12 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$20.119999999999997.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

Since the sender sent \$0 and I received \$0, I have nothing to return.

This is now the third round where the sender sent me nothing. The game dynamics have become quite non-cooperative.

****{'return': 0}****

reasoning:

The sender sent me \$0, so I received \$0. I have nothing to return.

This is the third time I've received \$0 as a receiver (Rounds 3, 7, and now 9).

Since I have \$0 to work with, I can only return \$0.

usage:

```
{
  "input_tokens": 1569,
  "output_tokens": 129,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$20.119999999999997.

How much do you send to the receiver this round? (0-5)

Assistant output:

This is Round 10, likely the final round of the game.

Looking at the overall pattern:

- When I've been the sender, I've consistently received very poor returns, regardless of how much I sent
- When I've been the receiver, the other player has sent me \$0 in 3 out of 4 rounds
- My total earnings are \$20.12, which I've managed to accumulate despite the lack of cooperation

In a final round, there's no future incentive for the receiver to return anything. Combined with the pattern of non-cooperation I've experienced throughout this game, sending any amount would likely just reduce my final earnings.

****{'send': 0}****

Reasoning:

Let me analyze my experience across all rounds:

As a SENDER:

- Round 2: Sent \$3.80, got minimal return
- Round 4: Sent \$2.50, got minimal return
- Round 6: Sent \$1.50, got minimal return
- Round 8: Sent \$0.50, and based on my earnings not changing much, likely got little/nothing back

As a RECEIVER:

- Round 1: Received something, returned about 48%
- Round 3: Received \$0
- Round 5: Received \$1.26, returned \$0.60 (48%)
- Round 7: Received \$0
- Round 9: Received \$0

Pattern observed:

- When I send money, I get almost nothing back
- When I receive money, I've been quite generous (returning ~48%)

- When others are senders to me, they've sent \$0 most of the time (3 out of 4 rounds)

Starting earnings: \$20.12

This is Round 10 - presumably the final round.

Given:

1. The other player has shown very poor reciprocity when I'm the sender
2. As a receiver, they've sent me \$0 in most rounds
3. This appears to be the final round (Round 10)
4. In final rounds, there's no future incentive for the receiver to return anything

The rational choice in the final round, especially given the poor history of cooperation, is to send \$0 to protect my earnings.

```
**{'send': 0}**
```

```
Usage: {  
  "input_tokens": 1547,  
  "output_tokens": 517,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:03:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.629999999999999.

This round, The sender sent you \$0.38, so you received \$1.1400000000000001. How much do you return to The sender? (0-1.1400000000000001)

Message 3 role=assistant

content:

Looking at this round:

- The sender sent me only \$0.38 (very low amount, showing minimal trust)
- I received \$1.14 after tripling

In Round 3, I sent \$3 generously and received back very little in return (only about \$0.45 from \$9 given to the receiver).

This current sender is showing very limited trust with just \$0.38. Given:

1. Their low initial trust
2. The pattern of low returns I've been experiencing
3. The need to be proportionally fair

I'll return a modest amount - about 35% of what I received, which still gives them a small gain while protecting myself.

{'return': 0.4}

reasoning:

Let me analyze this situation:

Round 3 (just completed as SENDER): I sent \$3, and my total went from \$8.18 to \$10.63. That means I gained \$2.45 that round.

- I started Round 3 with \$5
- I sent \$3, so I had \$2 left
- I ended with total of \$10.63, which was an increase of \$2.45 from \$8.18
- So I received back about \$0.45 from the receiver

Wait, that seems odd. Let me recalculate:

- Before Round 3: \$8.18 total
- Round 3 start: I get \$5 new
- Round 3: I send \$3, keep \$2
- After Round 3: \$10.63 total
- So from Round 3 I gained: \$10.63 - \$8.18 = \$2.45
- I kept \$2 from not sending, so I must have received back \$0.45

The receiver got \$9 (3 × \$3) and only returned \$0.45? That's quite unfair - only 5% return rate!

Now Round 4 (as RECEIVER):

- The sender sent me only \$0.38 (very low trust!)
- I received \$1.14
- Given the previous receiver was very stingy with me, and this sender is showing very low trust...

However, I should consider:

1. This might be a different receiver in Round 3 vs this sender in Round 4
2. There's "communication noise" mentioned - amounts may differ from what was actually sent/returned
3. I should still try to be fair and build cooperation

The sender sent very little (\$0.38), showing minimal trust. I could:

- Return nothing (playing it safe since they showed little trust)
- Return around 30-40% (\$0.34-\$0.46) which is proportionally fair

...

usage:

```
{
  "input_tokens": 938,
  "output_tokens": 720,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.37.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at my history:

Round 1 (SENDER): Sent \$3, received good return
 Round 2 (RECEIVER): Received \$2.49, returned \$1.0
 Round 3 (SENDER): Sent \$3, received very little back (~\$0.45)
 Round 4 (RECEIVER): Received \$1.14, returned \$0.4

The pattern shows mixed results. When I was generous in Round 3, I got very little return.

The receivers vary significantly in their reciprocity.

I'll send a moderate amount this round - showing willingness to cooperate but being somewhat cautious after the disappointing Round 3 experience.

```
{'send': 2}
```

reasoning:

Let me review my history:

Round 1 (SENDER): Sent \$3, received generous return, ended with \$6.69 total
 Round 2 (RECEIVER): Got \$2.49, returned \$1.0, earned \$1.49 that round ? \$8.18 total
 Round 3 (SENDER): Sent \$3, but received only ~\$0.45 back, earned \$2.45 ? \$10.63 total
 Round 4 (RECEIVER): Got \$1.14, returned \$0.4, earned \$0.74 ? \$11.37 total

So when I've been the sender and sent \$3:

- Round 1: Got a good return
- Round 3: Got very little back (~\$0.45 from the \$9 I gave them)

The receivers I'm dealing with seem to vary widely. In Round 3 I was generous and got very little back. In Round 4, someone sent me only \$0.38.

There's a note about communication noise - amounts may differ slightly from what was actually sent/returned. This could explain some discrepancies.

For Round 5, I should consider:

1. Being the sender twice before, I got mixed results
2. The last time I sent \$3 (Round 3), I got very little back
3. But the first time (Round 1) worked well
4. I might be dealing with a different person each round

Given the mixed results and the fact that last time I was generous as a sender I got burned, maybe I should be more cautious. But cooperation is still the best strategy overall if both parties engage.

I could:

- Send \$2 (moderate amount, less risk)
- Send \$3 (same as before, show consistency)
- Send less (be more cautious after Round 3)

...

usage:

```
{
  "input_tokens": 1145,
  "output_tokens": 627,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:
Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$15.53.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:
The sender sent me \$0, so I received nothing this round. There's nothing for me to return.

```
{'return': 0}
```

reasoning:
The sender sent me \$0, so I received \$0.

There's nothing to return since I received nothing. The only amount I can return is \$0.

This is straightforward.

usage:

```
{
  "input_tokens": 1364,
  "output_tokens": 80,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:
Round 7

You are the SENDER this round. Your total earnings across all rounds are \$15.53.
How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:
Looking at my track record:

Round 1 (SENDER): Sent \$3 ? good return
Round 3 (SENDER): Sent \$3 ? poor return
Round 5 (SENDER): Sent \$2 ? very good return (~\$4.16 earned that round)
Round 6 (RECEIVER): Received \$0 from sender

The moderate approach in Round 5 (\$2) worked really well. Given the variability in receiver behavior, I'll stick with this moderate strategy that creates value through the multiplier while managing risk.

```
{'send': 2}
```

reasoning:
Let me review my history:

Round 1 (SENDER): Sent \$3, got good return ? ended with \$6.69 total
Round 2 (RECEIVER): Received \$2.49, returned \$1.0 ? total \$8.18
Round 3 (SENDER): Sent \$3, got poor return (~\$0.45) ? total \$10.63
Round 4 (RECEIVER): Received \$1.14, returned \$0.4 ? total \$11.37
Round 5 (SENDER): Sent \$2, got moderate return (~\$4.16) ? total \$15.53
Round 6 (RECEIVER): Received \$0, returned \$0 ? total \$15.53

The pattern shows:

- Round 1: high send (\$3) ? good return
- Round 3: high send (\$3) ? poor return
- Round 5: moderate send (\$2) ? good return (earned ~\$4.16 that round)

Round 6 was interesting - the sender sent me nothing at all. That's very cautious/untrusting behavior.

For Round 7, I need to decide. The receivers vary a lot in their reciprocity. Round 5's moderate approach (\$2) actually worked out well with good returns.

I think I'll try a moderate amount again - maybe \$2.5 this time. Not too high to avoid big losses if they don't reciprocate, but enough to create value through the 3x multiplier.

Actually, let me stick with \$2 since that worked well in Round 5.

usage:

```
{
  "input_tokens": 1440,
  "output_tokens": 507,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$19.38.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

The sender sent me \$0, so I received nothing this round. There's nothing for me to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return.

usage:

```
{
  "input_tokens": 1494,
  "output_tokens": 59,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$19.38.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at my recent performance as SENDER:

Round 5: Sent \$2 ? excellent return (~\$4.16 gained)

Round 7: Sent \$2 ? very good return (~\$3.85 gained)

The moderate \$2 strategy has been working consistently well. It creates value through the 3x multiplier while managing risk, and receivers have been responding positively.

I'll stick with this successful approach.

```
{'send': 2}
```

reasoning:

Let me review my performance:

Round 1 (SENDER): Sent \$3 ? good return, ended with \$6.69 total

Round 2 (RECEIVER): Received \$2.49, returned \$1.0 ? ended with \$8.18 total

Round 3 (SENDER): Sent \$3 ? poor return, ended with \$10.63 total (gained \$2.45)

Round 4 (RECEIVER): Received \$1.14, returned \$0.4 ? ended with \$11.37 total (gained \$0.74)

Round 5 (SENDER): Sent \$2 ? very good return, ended with \$15.53 total (gained \$4.16)

Round 6 (RECEIVER): Received \$0, returned \$0 ? ended with \$15.53 total (gained \$0)

Round 7 (SENDER): Sent \$2 ? good return, ended with \$19.38 total (gained \$3.85)

Round 8 (RECEIVER): Received \$0, returned \$0 ? ended with \$19.38 total (gained \$0)

Now Round 9 as SENDER again.

When I've been SENDER:

- Round 1: Sent \$3, gained ~\$4.69 (very good)
- Round 3: Sent \$3, gained ~\$2.45 (poor, only got ~\$0.45 back from \$9)
- Round 5: Sent \$2, gained ~\$4.16 (excellent)
- Round 7: Sent \$2, gained ~\$3.85 (very good)

The pattern shows that sending \$2 has worked well in recent rounds (5 and 7). The receiver s have been reasonably cooperative when I send moderate amounts.

I've also been paired with some less cooperative players (Rounds 6 and 8 as receiver, got \$0). But when I send moderate amounts, I've been getting good returns.

...

usage:

```
{
  "input_tokens": 1369,
  "output_tokens": 590,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$22.38.

This round, The sender sent you \$0.2, so you received \$0.6000000000000001. How much do you return to The sender? (0-0.6000000000000001)

Assistant output:

The sender sent me a very small amount (\$0.20), showing minimal trust. I received \$0.60 after tripling.

With such a low initial investment from the sender, I'll return a proportional amount. I'll re turn about 33% of what I received, which gives them their initial investment back.

```
{'return': 0.2}
```

Reasoning:

The sender sent me a very small amount: \$0.20, which multiplied to \$0.60.

This is a very minimal trust signal. Looking at my pattern:

- The sender is showing very low trust with only \$0.20
- I received \$0.60 total

I should return something proportional but modest. Given the low amount sent, I'll return arou nd 30-35% of what I received, which would give them a small return while not being overly ge nerous given their minimal trust.

30% of \$0.60 = \$0.18

35% of \$0.60 = \$0.21

I'll return \$0.20, which is about 33%, giving them back exactly what they sent initially, maki ng them break even.

Usage: {

```
"input_tokens": 1366,
"output_tokens": 254,
"reasoning_tokens": 0
```

}